Dell PowerFlex 5.1.x and PowerFlex Manager 5.1.x

Software Administration Guide

August 2026
Rev. 1.1

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

The target audience for this document includes system administrators responsible for managing PowerFlex software through the PowerFlex Manager user interface or PowerFlex SCLI.

This document provides procedures for using Dell PowerFlex software through the PowerFlex Manager user interface or PowerFlex SCLI.

This Dell Technologies product uses an embedded Linux distribution that is licensed to Dell. This embedded Linux is only authorized to be used to operate the Dell product and only the Dell product.

Integrated PowerFlex management (non-KVM) was previously known as co-resident. Customer-provided infrastructure for PowerFlex management (Linux/KVM) was previously known as bring your own Linux. References in the documentation will be updated over time.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
device gateway target	DGWT
management virtual machine	MVM
metadata manager	MDM
PowerFlex data server	PDS
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

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Revision history

Table 2. Revisions

Date	Document revision	Description of changes
August 2026	1.1	Editorial changes
June 2026	1.0	Initial release

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Completing tasks in PowerFlex Manager

The following table specifies the location of tasks that you perform through PowerFlex Manager. The topic lists the tasks and corresponding location in the *Dell PowerFlex Manager User Guide*.

Table 3. Location of task in the Dell PowerFlex Manager User Guide

If you want to....	Go here in the Dell PowerFlex and PowerFlex Manager User Guide
Add a system data network in PowerFlex Manager	Adding a system data network
Add certificates for security	Security
Back up and restoring	Back up and restore
Back up PowerFlex Manager	Backing up
Change password	Changing your password
Complete the initial configuration and setup of your system	Initial configuration and setup
Configure system for events and alerts	Monitoring events and alerts
Create credentials for root and non-root users	Create credentials for root and non-root users
Define networks	Networking
Define network types	Defining a network
Deploy and manage system resources	Performing lifecycle operations for a resource group
Deploy a PowerFlex cluster using a CSV file	Deploy a PowerFlex cluster using a CSV topology file
Discover resources	Discover a resource
Download a configuration report	Exporting a configuration report for all resources and resource groups
Download a report that lists compliance details for resources	Exporting a compliance report for all resources
Edit the backup settings and detail	Editing backup settings and details
Generate a troubleshooting bundle	Generating a troubleshooting bundle
Get started with PowerFlex Manager	Getting started
Manage block storage	Managing block storage
Manage licenses	License management
Manage lifecycle of resource groups	Performing lifecycle operations for a resource group
Manage resource groups	Additional resource group management tasks
Manage resources	Additional resource management tasks
Manage templates	Additional template management tasks
Monitor software and firmware compliance	View compliance report
Monitor system resources and health	Displaying system information at a glance
Pause an inventory job	Pausing an inventory job
Perform software and hardware remediation	Viewing a compliance report for a resource
Protect your block environment	Protecting a block storage environment

Table 3. Location of task in the Dell PowerFlex Manager User Guide (continued)

If you want to....	Go here in the Dell PowerFlex and PowerFlex Manager User Guide
Remove a device	Remove a device from a storage node
Remove a system data network in PowerFlex Manager	Removing a system data network
Rename a system data network PowerFlex Manager	Renaming a system data network
Set up Dell Technologies connectivity services	Enabling Dell Technologies connectivity services
Update passwords	Updating passwords for system components
Upgrade software	Software upgrade
User management	Creating a user
Use templates to deploy your system	Preparing a template for a resource group deployment
View alerts	Viewing and acknowledging an alert
View events	Viewing an event
View jobs	Jobs
View the dashboard and interpret the graphs on the dashboard	Displaying system information at a glance
Work with repositories	Repositories

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Enable secure boot enforcement for PowerFlex compute-only nodes

To enable secure boot enforcement for PowerFlex compute-only nodes, you must meet the following prerequisites:

- Boot mode must be set to Unified Extensible Firmware Interface (UEFI) in **System BIOS Settings** > **Boot Settings**.

  NOTE: If the host is not in this mode, you may not be able to change it without reinstalling the operating system.

- The server must have the Trusted Platform Module (TPM) 2.0 installed.
- BIOS firmware must be at the required version for the specific PowerEdge model to support enabling secure boot. For more information, see the Dell Technologies support site.
- Linux distro must be deployed with a Secure-Boot-compatible Linux OS provided by the customer:
  - OS uses a shim-based boot flow
  - Kernel is signed by the OS vendor
- ESXi-based compute-only node is Secure-Boot-ready by default. Readiness is validated by using **secureBoot.py -c** before enabling secure boot.
- Secure boot must be enabled after the PowerFlex Manager deployment of the compute-only node.

Configure Dell PowerEdge iDRAC for secure boot

Use this procedure to configure iDRAC for secure boot.

Steps

1. Log in to the iDRAC web interface and click **Configuration** > **BIOS Settings** > **System Security**.
2. Set **TPM Security** to **On**.
3. Expand **TPM Advanced Settings** and set **TPM2 Algorithm Selection** to **SHA256**.
4. Set **Secure Boot** to **Enabled**.
5. From the **System Security Settings** page, click **Apply**.
6. Click **Apply** > **Reboot**.

Enable secure boot on ESXi-based compute-only nodes

Use the following section to enable secure boot on ESXi-based compute-only nodes for either partial support or full support.

Enable partial support in VMware ESXi

Secure boot must be enabled after PowerFlex Manager deploys the nodes. If it is enabled beforehand, deployments using PowerFlex Manager fail.

About this task

If TPM 2.0, secure boot, and SHA256 are enabled before VMware ESXi is installed (manually, not with PowerFlex Manager), this is automatically configured on the first reboot.

Partial Support: Trusted Boot with Attestation and passes with some partner VIBs.

- UEFI Secure Boot: Verifies bootloader and kernel modules at boot
- TPM Measurements: Stores boot hash measurements in TPM PCRs (used for attestation)
- TPM-Backed Encryption: VM, vSAN, and Core Dump
- VMware vCenter Attestation: Detects if the host booted in a tampered or untrusted state
- vTPM Support on VMs: VMs can be given a virtual TPM for in-guest security functions (also requires vCenter KMS server)

Steps

1. Run the validation script:
   ```
   /usr/lib/vmware/secureboot/bin/secureBoot.py -c
   ```
   If it passes, the following message appears: "Secure Boot CAN be enabled."

   If it fails, it lists unsigned VIBs. You must remove these before proceeding or the host experiences a purple screen on the next boot.
2. Enable SSH on the ESXi host and use any SSH client to connect to the ESXi host using the root user.
3. Verify security level:
   ```
   esxcli system settings encryption get
   ```
   The output should display:
   - Mode: **None**
   - Installed VIBs: **False**
   - Require Secure Boot: **False**
4. Enable TPM mode:
   ```
   esxcli system settings encryption set --mode=TPM --require-secure-boot=true
   ```
5. Reboot the host.
6. When the host is back online, verify the security level:
   ```
   esxcli system settings encryption get
   ```
   The output should display:
   - Mode: **TPM**
   - Installed VIBs: **False**
   - Secure Boot: **True**
7. Sync the configuration to the Bootbank:
   ```
   /bin/backup.sh 0
   ```

Enable full support in VMware ESXi

Use this procedure to enable full support in VMware ESXi.

Prerequisites

All vSphere Installation Bundles (VIBs) must be VMware-signed, or else Execution Control Lockdown fails. Signed VIB Enforcement ensures the following:

- VIBs are not altered
- Only VMware-signed VIBs can be installed
- Only signed VIBs can be loaded during ESXi boot

About this task

Full secure boot support in VMware ESXi includes all features of partial support.

Steps

1. Run the validation script:
   ```
   /usr/lib/vmware/secureboot/bin/secureBoot.py -c
   ```
   If it passes, the following message appears: "Secure Boot CAN be enabled."

   If it fails, it lists unsigned VIBs. You must remove these before proceeding or the host experiences a purple screen on the next boot.

2. Enable SSH on the ESXi host and use any SSH client to connect to the ESXi host using the root user.
3. Verify security level:

   ```
   esxcli system settings encryption get
   ```

   The output should display:
   - Mode: **TPM**
   - Installed VIBs: **False**
   - Require Secure Boot: **False**
4. If the output does not match the above, enable partial support following the instructions above before proceeding.
   a. To get the current settings, type:

      ```
      esxcli system settings encryption get
      ```

   b. To allow the kernel to accept VIB enforcement, type:

      ```
      esxcli system settings kernel set -s execInstalledOnly -v TRUE
      ```

   c. Power off the host and then power the host on (do not use reboot).
   d. To enable the VIB enforcement, type:

      ```
      esxcli system settings encryption set --require-exec-installed-only=T
      ```

   e. Reboot the node to enforce signed VIBs.
   f. Once the node is back online, verify the security level:

      ```
      esxcli system settings encryption get
      ```

   g. To sync the running configuration to the Bootbank, type:

      ```
      /bin/backup.sh 0
      ```

Backup keys and configurations

You must always backup recovery boot key and system configuration. Do not skip this step.

Steps

1. SSH to the VMware ESXi host as root.
2. Display the backup key and copy to a secure location off of the node:

   ```
   esxcli system settings encryption recovery list
   ```

   Copy the Recovery Key (second column) and paste into a text file to save for future recovery. The Recovery ID can be omitted.
3. Generate a host-level backup bundle:

   ```
   vim-cmd hostsvc/firmware/backup_config
   ```
4. Copy the web URL provided to download the backup bundle. Store this bundle in the same location as the recovery key backup text file.

   NOTE: The backup can take up to 30 seconds to complete and it is no longer accessible after 5 minutes.

Enable secure boot on Linux-based compute-only nodes

Use the following section to enable secure boot on Linux-based compute-only nodes.

Enable Linux

Steps

1. Verify secure boot is currently disabled:

   ```
   mokutil --sb-state
   ```

Expected output: `SecureBoot disabled`

2. Verify EFI boot path uses SHIM:

   `efibootmgr -v`

   Expected output: `BootXXXX* <OS> FILE (\EFI\<distro>\shim*.efi)`

   NOTE: The boot entry must already point to shim*.efi. Secure boot enablement does not modify EFI boot order, it only enforces signature validation.

Enable secure boot for Linux

Steps

1. SSH to your Linux host as root and validate secure boot is enabled on your machine:

   `mokutil --sb-stat`

   The output should have secure boot enabled.

2. If SDC is already installed, continue to step 4.
3. If SDC is not installed, install the SDC RPM. The installation should be successful, but the scini driver should fail to load. You should get the error message "scini service failed because the control process exited with error code".

   To check for details on the error:

   `systemctl status scini.service`

   `journalctl -xe`

   If you check dmesg, you should see: Loading of module with unavailable key is rejected
4. Change directory to `/bin/emc/scaleio/scini_sync/certs/`. In this directory are the SDC certificates.
5. Run the following command to confirm that they are valid and not expired:

   `openssl x509 -in <.pem file from directory> -noout -enddate | cut -d= -f2`

   The output is the expiration date.
6. If the certificate is valid, use the mokutil took to import the .der file. You are required to generate a password:

   `mokutil --import <.der file from directory> (Example: emc_scaleio2026.der)`

   There are two prompts to enter the password.
7. If the certificate supplied with the SDC package is expired, you may be supplied a certificate in .pem format which you must convert to .der format using the following command:

   `openssl x509 -in /usr/src/<file.pem> -outform DER -out /usr/src/<file.der>`

   If necessary, contact Dell Technologies Support to get the new signed SDC package and related certificates.
8. Reboot the host.
9. On boot, before the Linux OS boots, you must enter the **Perform MOK management** menu. Enter the MOK management and choose **Enroll MOK**.
   a. On the next screen, choose **View Key 0**.
   b. The next screen displays the key information. Press any key to go to the **Enroll key** page.
   c. On the next screen, enter the password you generated previously in step 6 and choose **Reboot**.
10. Log in to the host after reboot and run this command to validate the SDC is up and running:

    `systemctl status scini.service`
11. For kernel lockdown confirmation, type

    `cat /sys/kernel/security/lockdown`

    Expected output: `[integrity]`

    This indicates kernel lockdown is active due to secure boot.

Enable secure boot for Windows-based compute-only nodes

Use the following procedure to enable secure boot for Windows-based compute-only nodes.

Prerequisites

Ensure that you have configured Dell PowerEdge iDRAC for secure boot.

Steps

1. Log in to the Windows server.
2. Run this command to validate that secure boot is enabled on the server.
   a. Press **Win + R**.
   b. Type **`msinfo32`** and press **Enter**.
      BIOS mode must be **UEFI**.
      Secure Boot State must be **On**.
3. To validate from PowerShell, type **`Confirm-SecureBootUEFI`**
   The output should display **True**.

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Expanding PowerFlex storage

This section includes procedures to add PowerFlex nodes or add features to PowerFlex by editing an existing CSV file. It also contains a procedure on how to deploy or expand a PowerFlex cluster using REST API commands.

Expand PowerFlex using a CSV file

Before you expand a PowerFlex cluster, do the following:

1. Go to PowerFlex Software on the Dell Technologies support site and sign in.
2. Under **Keyword**, search for **`PowerFlex Software Installation Software`**.
3. Under **Category**, select **Full Release**.
4. Under **Version**, select your PowerFlex software version.
5. Click **Download** beside the PowerFlex software installation bundle name to download the `Software_Only_Complete_x.x.x_xxx.zip` file.
6. Extract the file when the download is complete.

If you use SSH keys, ensure that you use a single SSH key for the nodes when expanding a PowerFlex cluster using a CSV topology file.

While expanding nodes with SDTs, you may see errors listed in the **Events** tab. These errors are for the existing components, such as SDTs in the cluster.

Adding new components or features by editing the existing CSV file

To add components or new features to your system, use the existing CSV topology file for the PowerFlex system.

The CSV file that is used to deploy a system can be used later to add features or components to an existing PowerFlex system. The following scenarios are when you can use the CSV file at a later time:

- Add storage nodes.
- Add MDMs or tiebreakers. The PowerFlex installer adds them to the MDM cluster and configures them automatically.
- Add SDCs
- Add SDTs
- Configure storage pools
- Configure PowerFlex system features which can be configured using the CSV file, such as performance profiles and so on.

NOTE: You must use the saved copy of the CSV file that was used for the initial deployment to add nodes to the system.

To add and configure components and features, make a copy of the latest system CSV topology file. Add rows or columns, or edit existing rows and columns. For new features, you can copy columns from the latest complete CSV template, and paste them into the copy of your system's CSV file. Do the following to download example CSV files:

1. Log in to PowerFlex Manager.
2. Left-click the question mark (?) icon in the upper right.
3. Select **Getting Started**.
4. On the **Getting Started** page, click **Deploy With Installation File**.
5. In the **PowerFlex Installer** window, click **Next**.

   Under **Installation Topology**, you can download example CSV files.
6. To download the example CSV files, click the **Minimal** link (minimum fields) or the **Complete** link (all fields).

The following examples show how you can edit the CSV topology file for various system modification scenarios.

The following figure displays the CSV topology file before any modifications are made.

Username	Password	Operating	Core Cour	Socket Co	Is MDM/TE	MDM Mgm	MDM IPs	MDM Nam	Virtual IPs	Virtual IP	perfProfile	Protectior	StoragePo	Device Gr	Storage De	Storage De	is EC Servi	EC Service	EC IP Role	Is SDS	SDS Name	SDS All IPs	perfProfile	Fault Set
root	password	linux	128	2	Master	10.234.17	172.24.17	mdm0				pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	
root	password	linux	128	2	Slave	10.234.17	172.24.17	mdm1				pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	
root	password	linux	128	2	Slave	10.234.17	172.24.17	mdm2				pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	
root	password	linux	128	2	TB	10.234.17	172.24.17	mdm3				pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	
root	password	linux	128	2	TB	10.234.17	172.24.17	mdm4				pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	
root	password	linux	128	2		10.234.17	172.24.177.26,172.25.177.26,172.26.177.26,172.27.17					pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	
root	password	linux	128	2		10.234.17	172.24.177.27,172.25.177.27,172.26.177.27,172.27.17					pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	
root	password	linux	128	2		10.234.17	172.24.177.28,172.25.177.28,172.26.177.28,172.27.17					pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	
root	password	linux	128	2		10.234.17	172.24.177.41,172.25.177.41,172.26.177.41,172.27.17					pd1		dg1	/dev/disk/	dev1,dev2	yes	Node_10.2	172.24.17	no	SDS_10.23	172.24.17	High	

Storage Pool Configuration

Protectior	Device Gr	StoragePo	Data Layo	EC Storage	PMEM in u	Protection Scheme
pd1	dg1	sp1	EC	use_all	Yes	2+2

Device Group Configuration

Protectior	Device Gr	Media Typ	EC spare number of nodes
pd1	dg1	SSD	1
pd1	dg_pmem	PMEM	

Figure 1. An example CSV topology file

The following figure is updated to include Node182, Node183, and Node184, which displays as a new entry following the initial five nodes.

Username	Password	Operating	Is MDM/TE	MDM Mgm	MDM Ips	MDM Nam	perfProfile	Virtual IPs	Virtual IP	is EC Servi	EC Service	EC IP Role	Protectior	Device Gr	Storage De	Storage De	Is SDC	perfProfile	PMEM Dev	Is SDT	SDT Name	SDT Host I	SDT Storag	SDT All IPs
root	password	linux	Primary	172.24.10.	172.24.10.	MDM1	High	172.27.40.	ens161	Yes	Node177	172.24.10.	PD_EC1	DG1	/dev/nvme	EC177-dis	No	Normal	/dev/dax0	No	SDT177			
root	password	linux	Secondary	172.24.10.	172.24.10.	MDM2	High	172.27.40.	ens161	Yes	Node178	172.24.10.	PD_EC1	DG1	/dev/nvme	EC178-dis	No	Normal	/dev/dax0	No	SDT178			
root	password	linux	Secondary	172.24.10.	172.24.10.	MDM3	High	172.27.40.	ens161	Yes	Node179	172.24.10.	PD_EC1	DG1	/dev/nvme	EC179-dis	No	High	/dev/dax0	No	SDT179			
root	password	linux	TB	172.24.10.	172.24.10.	MDM4				Yes	Node180	172.24.10.	PD_EC1	DG1	/dev/nvme	EC180-dis	No	High	/dev/dax0	No	SDT180			
root	password	linux	TB	172.24.10.	172.24.10.	MDM5				Yes	Node181	172.24.10.	PD_EC1	DG1	/dev/nvme	EC181-dis	No	High	/dev/dax0	No	SDT181			
root	password	linux		172.24.10.	172.24.10.182					Yes	Node182	172.24.10.	PD_EC1	DG1			yes							
root	password	linux		172.24.10.	172.24.10.183					Yes	Node183	172.24.10.	PD_EC1	DG1			yes							
root	password	linux		172.24.10.	172.24.10.184					Yes	Node184	172.24.10.	PD_EC1	DG1			yes							

Storage Pool Configuration

Protectior	Device Gr	StoragePo	Data Layo	EC Storage	PMEM in u	Protection Scheme
PD_EC1	DG1	SP_EC1	EC	use_all	Yes	2+2

Device Group Configuration

Protectior	Device Gr	Media Typ	EC spare number of nodes
PD_EC1	DG1	SSD	1
PD_EC1	DG_pmem	PMEM	

Figure 2. Add an SDC to the CSV topology file

The following figure is updated to include Standby-Secondary and Standby-TB under the Is MDM/TB column.

Username	Password	Operating	Is MDM/TB	MDM Mgm	MDM Ips	MDM Nam	perfProfile	Virtual IPs	Virtual IP	is EC Servi	EC Service	EC IP Role	Protectior	Device Gr	Storage De	Storage De	Is SDC	perfProfile	PMEM Dev	Is SDT	SDT Name	SDT Host I	SDT Storag	SDT All IPs
root	password	linux	Primary	172.24.10.	172.24.10.	MDM1	High	172.27.40.	ens161	Yes	Node177	172.24.10.	PD_EC1	DG1	/dev/nvme	EC177-dis	No	Normal	/dev/dax0	No	SDT177			
root	password	linux	Secondary	172.24.10.	172.24.10.	MDM2	High	172.27.40.	ens161	Yes	Node178	172.24.10.	PD_EC1	DG1	/dev/nvme	EC178-dis	No	Normal	/dev/dax0	No	SDT178			
root	password	linux	Secondary	172.24.10.	172.24.10.	MDM3	High	172.27.40.	ens161	Yes	Node179	172.24.10.	PD_EC1	DG1	/dev/nvme	EC179-dis	No	High	/dev/dax0	No	SDT179			
root	password	linux	TB	172.24.10.	172.24.10.	MDM4				Yes	Node180	172.24.10.	PD_EC1	DG1	/dev/nvme	EC180-dis	No	High	/dev/dax0	No	SDT180			
root	password	linux	TB	172.24.10.	172.24.10.	MDM5				Yes	Node181	172.24.10.	PD_EC1	DG1	/dev/nvme	EC181-dis	No	High	/dev/dax0	No	SDT181			
root	password	linux	Standby-secondary	172.24.10.	172.24.10.182					Yes	Node182	172.24.10.	PD_EC1	DG1	/dev/nvme	EC182-dis	No							
root	password	linux	Standby-TB	172.24.10.	172.24.10.183					Yes	Node183	172.24.10.	PD_EC1	DG1	/dev/nvme	EC183-dis	No							

Storage Pool Configuration

Protectior	Device Gr	StoragePo	Data Layout	EC Storage	PMEM in u	Protection Scheme
PD_EC1	DG1	SP_EC1	EC	use_all	Yes	2+2

Device Group Configuration

Protectior	Device Gr	Media Typ	EC spare number of nodes
PD_EC1	DG1	SSD	1
PD_EC1	DG_pmem	PMEM	

Figure 3. Add an MDM and a tiebreaker to the CSV topology file

The following figure is updated to include the storage node (Node182), which displays as a new entry following the initial five nodes.

Username	Password	Operating	Is MDM/TB	MDM Mgm	MDM Ips	MDM Nam	perfProfile	Virtual IPs	Virtual IP	is EC Servi	EC Service	EC IP Role	Protectior	Device Gr	Storage De	Storage De	Is SDC	perfProfile	PMEM Dev	Is SDT	SDT Name	SDT Host I	SDT Storag	SDT All IPs
root	password	linux	Primary	172.24.10.	172.24.10.	MDM1	High	172.27.40.	ens161	Yes	Node177	172.24.10.	PD_EC1	DG1	/dev/nvme	EC177-dis	No	Normal	/dev/dax0	No	SDT177			
root	password	linux	Secondary	172.24.10.	172.24.10.	MDM2	High	172.27.40.	ens161	Yes	Node178	172.24.10.	PD_EC1	DG1	/dev/nvme	EC178-dis	No	Normal	/dev/dax0	No	SDT178			
root	password	linux	Secondary	172.24.10.	172.24.10.	MDM3	High	172.27.40.	ens161	Yes	Node179	172.24.10.	PD_EC1	DG1	/dev/nvme	EC179-dis	No	High	/dev/dax0	No	SDT179			
root	password	linux	TB	172.24.10.	172.24.10.	MDM4				Yes	Node180	172.24.10.	PD_EC1	DG1	/dev/nvme	EC180-dis	No	High	/dev/dax0	No	SDT180			
root	password	linux	TB	172.24.10.	172.24.10.	MDM5				Yes	Node181	172.24.10.	PD_EC1	DG1	/dev/nvme	EC181-dis	No	High	/dev/dax0	No	SDT181			
root	password	linux		172.24.10.	172.24.10.182					Yes	Node182	172.24.10.	PD_EC1	DG1	/dev/nvme	EC182-dis	No							

Storage Pool Configuration

Protectior	Device Gr	StoragePo	Data Layout	EC Storage	PMEM in u	Protection Scheme
PD_EC1	DG1	SP_EC1	EC	use_all	Yes	2+2

Device Group Configuration

Protectior	Device Gr	Media Typ	EC spare number of nodes
PD_EC1	DG1	SSD	1
PD_EC1	DG_pmem	PMEM	

Figure 4. Add a storage node to the CSV topology file

Deploy or expand a cluster using REST API commands

After you prepare the CSV topology file, use the REST API to parse it and upload the configuration to PowerFlex Manager. Then, deploy the block storage resources using the REST API commands. The following procedure is only recommended for advanced users familiar with working with the REST API.

Prerequisites

- Ensure that you have prepared the CSV topology file.
- Ensure that you know the load balancer IP address and user credentials of PowerFlex Manager. All the steps that are performed must use the load balancer IP address. The load balancer IP address is the same IP address that you are using to access PowerFlex Manager.
- Ensure that you have the software packages of the PowerFlex core components that you want to deploy, and the ActiveMQ package.
- See the Dell Technologies Developer portal for information about working with the PowerFlex Manager REST API.

About this task

The following procedure provides examples using cURL.

Steps

1. Get the configuration from the CSV file by parsing it using the PowerFlex REST API. Use the API call `POST /im/types/Configuration/instances/actions/parseFromCSV`, where the body is the CSV file reference.

   For example:

   ```
   curl -v -k -X POST -H "Content-Type:application/json" -H "Authorization: Bearer
   ${token}" -F "file=<PATH_TO_CSV_FILE>" https://<IP_ADDRESS>/im/types/Configuration/
   instances/actions/parseFromCSV
   ```

   Where: *{token}* is the Keycloak token, *<PATH_TO_CSV_FILE>* is the path to the CSV topology file, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.

   The command returns a JSON file containing the required cluster configuration. The installation command requires this file in order to install resources.
2. Save the JSON file.
3. To add packages, copy the installation packages of the required core components, including the ActiveMQ RPM, to `/usr/local/tomcat/temp/scaleio`.
4. Open a shell and run the following command:

   ```
   kubectl get pods -n powerflex | grep block-legacy
   ```

5. Make a note of the block-legacy-gateway ID that is returned in the first line of the command output. In the following example, the ID is `block-legacy-gateway-0`:

   ```
   admin@hostname:~> kubectl get pods -n powerflex | grep block-legacy
   block-legacy-gateway-0                                2/2     Running     0
     76d
   block-legacy-gateway-1                                2/2     Running     0
     76d
   block-legacy-gateway-mds-0                            1/1     Running     0
     76d
   ```

6. To log in to the pod for the legacy gateway, run:

   ```
   kubectl exec --stdin --tty <ID> -n powerflex -- /bin/sh
   ```

   where *<ID>* is the block-legacy-gateway ID obtained in the previous step.

7. To update the open SSH client and scp, run:

   ```
   apt-get update apt-get install openssh-client scp
   root@<RPM_MACHINE_IPS>:<RPMS_LOCATION>* temp/scaleio
   ```

Where *<RPM_MACHINE_IPS>* is the IP address where the packages are saved, and *<RPMS_LOCATION>* is the path to the location of the packages.

8. Install block storage resources using the PowerFlex REST API. Use the API call `POST /im/types/Configuration/actions/install`, where the body is the cluster configuration JSON file that was returned from parsing the CSV file.

   If the parameter *systemId* (type string) is added to the configuration, the cluster is created with the given ID. Otherwise, the gateway generates an ID.

   Optional parameters include:

   - `noUpload`—Skip upload package phase (true or false, default if false)
   - `noInstall`—Skip install package phase (true or false, default if false)
   - `noConfigure`—Skip configure phase and repository files (true or false, default if false)
   - `noSecurityBootstrap`—Skip security bootstrap (part of install phase) (true or false, default if true)
   - `noLinuxDevValidation` (true or false, default if true)
   - `storagenodeInstances` (number, optional)
   - `extend`—extend an existing system (true or false, default if false)
   - `globalZeroPadPolicy` (true or false, default if true)
   - `forceDeviceTakeover`—Clean devices before installing storage nodes (without it, installation will fail when it runs after uninstall; must not be used in production) (true or false, optional)

   For example:

   ```
   curl -v -s -k -i -X POST -H "Content-Type:application/json" -H
   "Authorization: Bearer ${token}" "https://<IP_ADDRESS>/im/types/Configuration/actions/
   install?noSecurityBootstrap=false&forceDeviceTakeover=true" -d @<PATH>/config.json
   ```

   Where: *{token}* is the Keycloak token, *<IP_ADDRESS>* is the PowerFlex Manager IP address, and *<PATH>* is the path to the JSON configuration file that was output in step 1.

9. Installation is asynchronous and consists of four stages: query, upload, install, and configure. Manually monitor and move from one phase to the next, using the commands `queryPhaseState` and `moveToNextPhase`. You can also monitor the progress of each stage by viewing events in PowerFlex Manager.

   The following is an example of an installation event:

   ```
   COMMAND_STARTED_ON_OPERATION_INSTALL_AND_PHASE_QUERY
   ```

10. After installation is complete, clear the gateway state using the API call `POST im/types/Command/instances/actions/clear`.

    For example:

    ```
    curl -v -s -k -i -X POST -H "Content-Type:application/json" -H "Authorization: Bearer
    $<token>" "https://<IP_ADDRESS>/im/types/Command/instances/actions/clear"
    ```

    Where: *<token>* is the Keycloak token, and *<IP_ADDRESS>* is the PowerFlex Manager IP address.

6

Maintaining nodes in an integrated PowerFlex management (non-KVM) deployment

In an integrated PowerFlex management (non-KVM) deployment, you can move a node in and out of maintenance mode so you can perform administrative maintenance tasks.

Integrated PowerFlex management (non-KVM) was previously known as co-resident. Customer-provided infrastructure for PowerFlex management (Linux/KVM) was previously known as bring your own Linux. References in the documentation will be updated over time.

Determine the primary MDM

Use the following procedure to determine if you are working with a primary metadata manager (MDM).

Prerequisites

Establish an SSH connection to the node.

Steps

1. To determine if the node is a primary MDM, run:

   ```
   scli --query_cluster
   ```

2. Check the output of the previous command for cluster information.
   - If the command does not return cluster information, you can perform maintenance on the node.
   - If the command returns cluster information, you must switch ownership to the secondary MDM.

   To log in to the PowerFlex cluster, run:

   ```
   scli --login --username admin --management_system_ip <powerflex_manager_IP>
   ```

   To switch cluster ownership, run:

   ```
   scli --switch_mdm_ownership --new_primary_mdm_name <mdm_name>
   ```

Related information

Prepare a PowerFlex management node for maintenance

Prepare a PowerFlex management node for maintenance

You must terminate or move the pods running on the PowerFlex node before updating, upgrading, or rebooting a PowerFlex management node.

Prerequisites

Ensure that ownership is switched from the primary MDM to the secondary MDM before any maintenance activities are performed on the node.

About this task

Before updating, upgrading, or rebooting a PowerFlex management node, the pods running on the PowerFlex node must be properly terminated or moved.

Steps

1. Establish an SSH connection to the PowerFlex controller node.
2. Label the node for maintenance mode.
   a. To display existing node status and role type, run the following command:

   ```
   kubectl get nodes
   ```

   ```
   Example output:
   NAME            STATUS   ROLES                         AGE   VERSION
   node1           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
   node2           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
   node3           Ready    control-plane,etcd,master   24h   v1.25.3+rke2r1
   ```

   b. Label the node for maintenance by running the following command:

   ```
   kubectl label node <NAME> cmo.maintenance.mode=true
   ```

   ```
   kubectl label node node1 cmo.maintenance.mode=true
   ```

3. Verify if the node is a primary MDM by running the following command:

   ```
   scli --query_cluster
   ```

   Check the command output for cluster information. If the command returns cluster information, run the following commands to switch ownership to the secondary MDM. Otherwise, continue to step 4.

   a. Log in to the PowerFlex cluster:

   ```
   scli --login --username admin --management_system_ip <powerflex_manager_IP>
   ```

   b. Run this command to switch cluster ownership:

   ```
   scli --switch_mdm_ownership --new_primary_mdm_name <mdm_name>
   ```

4. Drain the pods that are running on the node with the following command:

   ```
   kubectl drain <NAME> --ignore-daemonsets --delete-emptydir-data
   ```

   NOTE: The command may take more than a minute to complete, as it is not finished when the prompt returns. The process continues in the background, and you must wait four to five minutes before proceeding.

Next steps

Enter a storage node into maintenance mode. See **Related information**.

Related information

Determine the primary MDM
Enter a storage node into maintenance mode

Enter a storage node into maintenance mode

You must use PowerFlex Manager to enter the storage node into maintenance mode.

Prerequisites

Use the following procedure for a node in an integrated PowerFlex management (non-KVM) deployment.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Lifecycle** > **Storage Nodes**.
3. From the list of storage nodes, select the check box beside the storage node that you want to enter into maintenance mode and click **More Actions** > **Enter Maintenance Mode**.
4. In the **Enter Storage Node into Maintenance Mode** dialog box, verify that you are selecting the wanted storage node, and click **Enter Maintenance Mode**.
5. Verify that the operation is finished and is successful, and click **Dismiss.**

Results

The node is now ready for maintenance. You can update the node and reboot it. When maintenance is complete, return the node to operation.

Related information

Prepare a PowerFlex management node for maintenance
Exit a management node from maintenance mode
Exit a storage node from maintenance mode

Exit a management node from maintenance mode

To return the node to operation after maintenance, you must wait for the node to rejoin the cluster before exiting the node from maintenance mode.

Steps

1. Establish an SSH connection to the node.
2. Wait for the node to rejoin the cluster. From a different node that is part of the integrated PowerFlex management (non-KVM) deployment, run the following command to return the node to service:

   ```
   kubectl get nodes
   ```

   Continue to wait if the status is anything other than `Ready, SchedulingDisabled`.
3. To bring the node back into service, run the following command:

   ```
   kubectl uncordon <NAME>
   ```

   <NAME> is the name of the node.

4. Check the status by running the following command:

```
kubectl get nodes
```

   Wait for the status to report `Ready` before continuing.

5. Optional: If you were also performing maintenance on a storage node, do the following:
   a. Log in to PowerFlex Manager.
   b. Go to **Lifecycle** > **Storage Nodes** and select the checkbox of the storage node you are working on.
   c. Click **More Actions** and select **Exit Maintenance Mode**.
   d. Wait until the PowerFlex management platform status indicator light goes green before continuing.
6. To remove the maintenance mode label from the management node, run the following command:

```
kubectl label node <NAME> cmo.maintenance.mode-
```

   For example:

```
kubectl label node node1 cmo.maintenance.mode-
```

7. In PowerFlex Manager, go to **Monitoring** > **Management Diagnostics** > **Nodes**.

   The **Nodes** page opens which displays a list of PowerFlex management platform nodes.

   - You can click **Add filter** and enter case-sensitive free text to control the list of nodes displayed.

8. Select the node that you want to view.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

9. Wait until the PowerFlex management platform status indicators change to green before moving on to the next node.

Related information

Enter a storage node into maintenance mode
Post-maintenance status check for PowerFlex Manager version 5.1

Exit a storage node from maintenance mode

To return the node to operation after maintenance, you must wait for the node to rejoin the cluster before exiting the node from maintenance mode.

Steps

1. Log in to PowerFlex Manager.
2. Go to **Lifecycle** > **Storage Nodes** and select the checkbox of the storage node you are working on.
3. Click **More Actions** and select **Exit Maintenance Mode**.
4. Wait until the PowerFlex management platform status indicators change to green before moving on to the next node.

Related information

Enter a storage node into maintenance mode
Post-maintenance status check for PowerFlex Manager version 5.1

Post-maintenance status check for PowerFlex Manager version 5.1

You must verify the readiness of the nodes, pods, and PowerFlex Manager database after maintenance activities.

Steps

1. Log in to PowerFlex Manager.
2. Click on **Monitoring** > **Management Diagnostics**.
3. Verify green status on all panels.

Related information

Exit a management node from maintenance mode
Exit a storage node from maintenance mode

Shutting down or rebooting non-PowerFlex Manager nodes

You can shut down or reboot non-PowerFlex Manager nodes, and then return the nodes to an operational state.

Shut down or reboot a non-PowerFlex Manager node

Prepare the node for an update or maintenance operation by entering the node into maintenance mode and shutting down or rebooting the node in a graceful manner.

Prerequisites

To retrieve the MDM management IP address through the PowerFlex Manager user interface, you must first log in using the username and password. Once logged in, complete the following steps to retrieve the MDM management IP address:

1. Log in to PowerFlex Manager.
2. On the menu bar, go to **Lifecycle** > **Resources**.
3. Select the **PowerFlex system** resource and click **View Details** to view the MDM management IP address.

About this task

CAUTION:

- **When performing tasks on a node that require it to be shut down or rebooted, you must gracefully shut down or reboot the node. Do not force the shut down or reboot operations.**
- **Operating system updates and other maintenance activities, like part replacement, require that you gracefully shut down or reboot a node.**
- **Do not use `-f` in the shutdown or reboot command.**

Steps

1. When shutting down or rebooting a node that is a primary MDM (manager), it is recommended that you manually switch MDM ownership to a different node:
   a. Log in to the primary MDM using the following command:

      ```
      scli --login --username admin --management_system_ip <powerflex_manager_IP>
      ```

   b. To query the cluster from the PowerFlex CLI (SCLI), run:

      ```
      scli --query_cluster
      ```

c. If the IP addresses of the node are in the `--query_cluster` output, the faulty node has a role of either MDM or tiebreaker, in addition to its storage node role.

   If the IP address of the node is in the primary MDM role, a switch-over action is required.

d. To switch MDM ownership to a different node, run:

```
scli --switch_mdm_ownership (--new_primary_mdm_id <ID> | --new_primary_mdm_ip <IP>
| --new_primary_mdm_name <NAME>)
```

   The node remains in the cluster. The cluster will be in degraded mode after it is powered off, until the faulty component or patch operation in the node is fixed and the node is powered back on.

e. To verify that the cluster status shows that the node is not the primary MDM anymore, run:

```
scli --query_cluster
```

   Output similar to the following displays, with the relevant node configuration and IP addresses for your deployment:

```
Cluster:
    Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
    Virtual IP Addresses: 9.20.10.100, 9.20.110.100
Primary MDM:
    ID: 0x775afb2a65ef1f02
        IP Addresses: 9.20.10.104, 9.20.110.104, Management IP Addresses:
10.136.215.239, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Version: 2.0.13000
Secondary MDMs:
    ID: 0x5b2e9f273b7af9b0
        IP Addresses: 9.20.10.105, 9.20.110.105, Management IP Addresses:
10.136.215.223, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Status: Normal, Version: 2.0.13000
    ID: 0x5828f65b15e778f1
        IP Addresses: 9.20.10.102, 9.20.110.102, Management IP Addresses:
10.136.215.232, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Status: Normal, Version: 2.0.13000
Tiebreakers:
    ID: 0x6618e0b804644ca4
        IP Addresses: 9.20.10.101, 9.20.110.101, Port: 9011
        Status: Normal, Version: 2.0.13000
    ID: 0x12534ccb3d28fee3
        IP Addresses: 9.20.10.103, 9.20.110.103, Port: 9011
        Status: Normal, Version: 2.0.13000
```

   In the example output, the primary MDM IP addresses are:

```
IP Addresses: 9.20.10.104, 9.20.110.104, Management IP Addresses: 10.136.215.239
```

   The secondary IP addresses are:

```
IP Addresses: 9.20.10.105, 9.20.110.105, Management IP Addresses: 10.136.215.223
IP Addresses: 9.20.10.102, 9.20.110.102, Management IP Addresses: 10.136.215.232
```

2. Log in to PowerFlex Manager as an admin user.
3. On the menu bar, click **Lifecycle** > **Storage Nodes**.
4. Select the relevant storage node and click **More Actions** > **Enter Maintenance Mode**.
5. Click **Enter Maintenance Mode**.
6. In the confirmation message dialog box, click **Dismiss**.
7. If you are applying a patch:
   a. Run the patch.
   b. Reboot the node, if necessary.

Return the node to an operational state

After a maintenance activity, you must return the node to an operational state.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Lifecycle** > **Storage Nodes**.
3. Select the checkbox for the node in maintenance mode, go to **More Actions**, and select **Exit Maintenance Mode**.
4. Wait until the storage node state displays as **Healthy**.
5. Wait for the rebuild/rebalance operations to complete.
   The node is now operational.

Running scripts on hosts

You can run scripts on servers hosting PowerFlex MDM or storage node components. You can use the scripts to update operating systems or firmware. This feature is supported only on Linux-based bare-metal nodes or virtual nodes.

The PowerFlex Installer can run a user-provided script on a node where PowerFlex is deployed. You can use this feature for any purpose that is external to the PowerFlex system, such as running CLI commands or updating an operating system. This feature enables you to run scripts in a secure fashion while maintaining data integrity.

The PowerFlex Installer orchestrates running the script, and ensures that storage nodes are placed in maintenance mode to protect data during the process. Parallel script execution is only permitted on storage nodes that are in different protection domains. After a script is run on a storage node, the scripts exits the storage node from maintenance mode.

Optionally, you can set the servers to reboot after running the script. You can also run a verification script either after the initial script is run or after the server reboots.

This feature includes the following system validation before running a script:

- The script was uploaded to the node.
- There is no failed storage capacity.
- There is sufficient spare storage capacity.
- The MDM cluster is in a valid state.

Run the script on one host, and prioritize the hosts in the following manner:

1. Storage node-only hosts, each time on a single storage node, unless the option `In parallel on different Protection Domains` is enabled.
2. Tiebreaker
3. MDMs

> **NOTE:** The script cannot run on a primary MDM. A switch-over MDM command is used before the script runs on a primary MDM. The script cannot be run in parallel on multiple MDMs, or tiebreakers.

Running the script proceeds as follows:

1. The node enters maintenance mode.
2. The script runs on the host.
3. The host reboots, if required.
4. The validation script runs on the node, if required.
5. The node exits maintenance mode.

Run a script on a host

Use this method to run a script on all or some of the Linux-based nodes with an optional reboot and an optional verification script.

> **CAUTION: PowerFlex Manager is required to run a script on a host. A target script is run without validation or verification.**

The script files must be stored in the folder node folder: `/opt/emc/scaleio/lia/bin`. The script file used to run the script on the host operation must be named as `patch_script`.

If verification is required, a script to verify the work can be written and named `verification_script`.

The filenames are hard coded and cannot be changed from: `patch_script` and `verification_script`. The scripts are required to have these names. Alternatively, they can be uploaded through PowerFlex Manager to the PowerFlex node.

Use the `kubectl cp` command to copy the script file from any of the PowerFlex management platform hosts or VMs to the block-legacy-gateway pods.

For example:

```
kubectl cp -n powerflex patch_script block-legacy-gateway-0:/usr/local/tomcat/temp/
```

```
kubectl cp -n powerflex patch_script block-legacy-gateway-1:/usr/local/tomcat/temp/
```

The scripts can be either taken from a gateway local folder or downloaded from an HTTP or HTTPS share.

A list of `storagenodeIds` or `mdmIds` can be provided to explicitly choose the PowerFlex nodes to run on.

API call

```
/im/types/Configuration/actions/liaRunOsPatching
```

NOTE: Before sending the `liaRunOsPatching` call, log in to PowerFlex and get the system configuration. For more information, see the example commands and workflow below.

Required parameters

One of the following parameters is required:

pdIds

Run on all PowerFlex nodes that are part of the following protection domains (PD Ids), in decimal format.

storagenodeIds

Run on all storage nodes listed by Ids, in decimal format.

mdmIds

Run on all MDMs listed by Ids, in decimal format.

executeOnAllstoragenodes

Run on all storage nodes (true/false)

NOTE: All nodes running PowerFlex management platform processes are skipped for reboots. The action must be performed manually.

executeOnAllMdms

Run on all MDMs (true/false)

Optional parameters

The following parameters are optional:

isRebootRequired

Indicates if each node reboot is required after running the patch script (values: true/false)

NOTE: Nodes that are running PowerFlex management platform processes are skipped for reboots. The action must be performed manually.

isVerificationScriptRequired

Indicates if the verification script is run on each node (values: true/false)

isRunningInParallelOnPds

Indicates if the operation is run in a parallel way on nodes that belong to different PDs (values: true/false)

isStopProcessingOnScriptFailure

Indicates if the entire operation must be stopped if there is a script failure (values: true/false

timeout_sec

Indicates timeout value for running the patch script in milliseconds

isUploadFileNeeded

Indicates if the gateway upload scripts to the PowerFlex nodes (values: true/false)

The following fields are relevant when `isUploadFileNeeded` is true:

patchScriptFilePath

Either the local folder name on the block-legacy-gateway pods or an http/https URL of the patch script

verificationScriptFilePath

Either the local folder name on the block-legacy-gateway pods or an http/https URL of the verification script

maintenanceModeType

Maintenance Mode type (values: IMM)

verificationScriptTimeoutSec

Verification script timeout in seconds

rebootTimeoutSec

Node reboot timeout in seconds

Example commands and workflow

The following are the example commands that are used when running a script on a host:

1. Obtain an access token from the PowerFlex Manager instance. The easiest method is to create a shell script that can be sourced to add the proper variables to the user environment:

```
INGRESS_IP=<powerflex_manager_IP>
INGRESS_USER=<powerflex_manager_user>
INGRESS_PASSWORD=<powerflex_manager_password>
```

   a. Get a JWT token with `POST /rest/auth/login`:

```
TOKEN=$(curl -s -k --location --request POST "https://${INGRESS_IP}/rest/auth/
login" --header "Accept: application/json" --header "Content-Type: application/
json" --data "{\"username\": \"${INGRESS_USER}\",\"password\": \"${INGRESS_PASSWORD}
\"}")
```

   b. Parse out the access token which is used to call the API and is valid for five minutes by default:

```
ACCESS_TOKEN=$(echo "${TOKEN}" | jq -r .access_token)
```

   c. Parse out the refresh token which can be used to get a new JWT token if the access token has expired. It is valid for 30 minutes by default:

```
REFRESH_TOKEN=$(echo "${TOKEN}" | jq -r .refresh_token)
```

   NOTE: The expiration time for access token is five minutes. If required, the above file can be sourced to refresh all variables.

2. Get a system configuration JSON file, and use it as the payload of the patch command. You must replace `liaPassword` and `mdmPassword` manually from null to some string.
   a. To create and save a JSON file like in the following example, replace the MDM addresses, MDM user, and MDM password with the appropriate values:

```
{ "mdmIps":["<MDM IP-1>","<MDM-IP2>"], "mdmUser":"<mdm user>", "mdmPassword":"<mdm
password>", "securityConfiguration": { "allowNonSecureCommunicationWithMdm":"true",
```

```
"allowNonSecureCommunicationWithLia":"true", "disableNonMgmtComponentsAuth":"false"
} }
```

b. Insert the output of this command (with fixed passwords) into the configuration JSON file:

```
curl -s -X POST -k -H "Content-Type: application/json" --data <json file>
-H "Authorization: Bearer ${ACCESS_TOKEN}" https://<powerflex manager IP>/im/types/
Configuration/instances > config.json
```

3. To load the configuration.json file, run:

```
curl -v -k -c cookiefile -X -i POST -H "Content-Type:application/json" -H
"Authorization:
Bearer ${ACCESS_TOKEN}" "https://<powerflex manager IP>/im/types/Configuration/
actions/liaRunOsPatching?executeOnAllStoragenodes=true
&isRebootRequired=true&isPatchScriptRequired=true&isVerificationScriptRequired=true&pa
tchScriptFilePath=”https://
<ipaddress>/patch_script”&verificationScriptFilePath=”https://<ip-address>/
verification_script”&maintenanceModeType=IMM&rebootTimeoutSec=600" -d @config.json
```

NOTE: The `cookiefile` stores session cookies which keep the command flow directed to the same gateway pod.

4. To query the command state, run:

```
curl -k -b cookiefile -X GET -H "Content-Type: application/json" -d @config.json
-H "Authorization: Bearer $ACCESS_TOKEN" "https://10.234.89.250/im/types/ProcessPhase/
actions/queryPhaseState" | jq
{
  "phaseStatus": "idle",
  "phase": "idle",
  "numberOfRunningCommands": 0,
  "numberOfPendingCommands": 0,
  "numberOfCompletedCommands": 0,
  "numberOfAbortedCommands": 0,
  "numberOfFailedCommands": 0,
  "failedCommands": []
}
```

You can also run:

```
{ "phaseStatus": "running", "phase": "execute", "numberOfRunningCommands":
1, "numberOfPendingCommands": 1, "numberOfCompletedCommands": 35,
"numberOfAbortedCommands": 0, "numberOfFailedCommands": 0, "failedCommands": [] }
```

5. To move to the next phase, run:

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d ''
-H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250//im/types/ProcessPhase/
actions/moveToNextPhase
```

6. To cancel the phase, run:

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d
'' -H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/Command/
instances/actions/abort
```

7. To clear the phase, run:

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d
'' -H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/Command/
instances/actions/clear
```

8. To move to idle phase, run:

```
curl -k -b cookiefile -X POST -H "Content-Type: application/json" -d ''
-H "Authorization: Bearer $ACCESS_TOKEN" https://10.234.89.250/im/types/ProcessPhase/
actions/moveToIdlePhase
```

Logs

The gateway pod log locations are:

- `/usr/local/tomcat/logs/scaleio.log`
- `/usr/local/tomcat/logs/scaleio-trace.log`

The LIA log location is:

- `/opt/emc/scaleio/lia/logs/trc.x`

NOTE: There is a special switch to keep the script in the node when troubleshooting or testing:

1. Edit the file `/usr/local/tomcat/webapps/ROOT/WEB-INF/classes/gatewayInternal.properties` .
2. Find the field: `"ospatching.delete.scripts=false"`
3. Change to true for troubleshooting. The default is false.

Fully powering off or fully powering on nodes

You can fully power off or fully power on nodes in an integrated PowerFlex management (non-KVM) deployment. These actions are not part of routine maintenance.

The power off procedure FULLY powers off a PowerFlex system. When you complete the steps in *Power off nodes in an integrated PowerFlex management (non-KVM) deployment*, you lose all access to your PowerFlex cluster.

The power on procedure FULLY powers on a PowerFlex system. Complete the steps in *Power on nodes in an integrated PowerFlex management (non-KVM) deployment* to power on a PowerFlex cluster that is fully powered off.

Power off nodes in an integrated PowerFlex management (non-KVM) deployment

Use the following procedure to fully power off the nodes in an integrated PowerFlex management (non-KVM) deployment.

About this task

CAUTION: When you fully power off the PowerFlex system, you lose all access to your PowerFlex cluster.

Steps

1. Stop all external workloads. Dell Technologies recommends that you unmount the volumes for the SDCs.
2. Log in to any PowerFlex Manager node using SSH.
3. Stop the PowerFlex Manager database.

   NOTE: Once the PowerFlex Manager database is stopped, the PowerFlex Manager user interface is unavailable, and you cannot authenticate to the MDM cluster.

   a. To gather the IP address of the database service, run these commands:

   ```
   export NAMESPACE=powerflex
   ```

   ```
   export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
   ```

   b. To shut down the database pods and service, run this command:

   ```
   curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdown"
   ```

   c. To check the shutdown status, run this command:

   ```
   curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdownstatus"
   ```

d. Continue to run the shutdown status command until the command returns a verified status.

The following example output shows a verified status:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"startup_status":"verified"}
```

4. Power off all nodes.

NOTE: Nodes running PowerFlex management platform processes may take longer to shut down than simple storage nodes.

Power on nodes in an integrated PowerFlex management (non-KVM) deployment

Use the following procedure to fully power on the nodes in an integrated PowerFlex management (non-KVM) deployment.

About this task

Fully power on the PowerFlex system.

Steps

1. Power on the storage nodes.
2. Log in to the primary MDM using SSH.
3. To validate whether the cluster status is normal, run:

```
scli --query_cluster
```

4. Log in to the nodes.
5. If you have an SDC installed, from each node, check for the status of exported volumes using the `lsblk` command until the partitions are displayed under the `scinia` device.
   If the partitions are not visible, but the cluster reports as healthy, use the `partprobe` command to rescan the attached block devices.
6. To check the status of the rke2-server process on the nodes, run:

```
systemctl status rke2-server
```

- If the status is active, go to the next step.
- If the status is shown as activating, repeat the status check until active.
- If the status is failed, to attempt to start the service, run:

```
systemctl start rke2-server
```

7. Once rke2-server is running, ensure the PowerFlex management platform nodes are in a ready state by running:

```
kubectl get nodes
```

If a message displays similar to the message below, wait for a few minutes, and verify again:

```
NAME    STATUS    ROLES                         AGE     VERSION
mvm1    Ready     control-plane,etcd,master     2d4h    v1.30.8+rke2r1
mvm2    Ready     control-plane,etcd,master     2d6h    v1.30.8+rke2r1
mvm3    Ready     control-plane,etcd,master     2d6h    v1.30.8+rke2r1
```

When the three MVM nodes are ready, you must restore the PowerFlex Manager database.

8. To set up the alias and default context, run:

```
alias k="kubectl -n $(kubectl get pods -A | grep -m 1 -E 'platform|pgo|helmrepo' |
cut -d' ' -f1)" kubectl config set-context default --namespace=$(kubectl get pods -A
| grep -m 1 -E 'platform|pgo|helmrepo|docker' | cut -d' ' -f1)
```

9. To restart the database, run:

```
k patch $(k get postgrescluster -o name) --type merge --patch '{"spec":{"shutdown":
false}}'
```

10. In PowerFlex Manager, go to **Monitoring** > **Management Diagnostics** > **Pods**.

    The **Pods** page opens which displays a list of PowerFlex management platform pods.

    - You can click **Add filter** and enter case-sensitive free text to control the list of pods displayed.
    - You can add additional filters by selecting **Status**, **Namespace**, or **Node Name**.

    NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

11. Select the pod that you want to view.
12. Wait for the pod status indicator to change to green.

    CAUTION: If the pod status remains red or the PowerFlex Manager user interface is not accessible after 15 to 20 minutes, contact Dell Technologies Support for further assistance.

7

Managing CloudLink

Administrators can use PowerFlex Manager to manage a CloudLink environment. Administrators can configure syslog servers, configure logging options, and use CloudLink Center to monitor CloudLink.

Licensing CloudLink

CloudLink licensing is not required for CloudLink version 8.3 and higher.

Inventory CloudLink Center VMs

If the CloudLink environment is managed by PowerFlex Manager, you must inventory the CloudLink Center VMs.

Steps

1. Log in to PowerFlex Manager.
2. Go to **Lifecycle** > **Resources**.
3. In the **Resources** page, select the CloudLink Center VMs and click **Run Inventory**.

Monitoring CloudLink health status

You cannot monitor CloudLink health status in PowerFlex Manager, because PowerFlex Manager does not have access to CloudLink health monitoring information. Since PowerFlex Manager is not able to determine the health status of CloudLink, it is not updated.

NOTE: PowerFlex Manager displays an expected **unknown** health status for CloudLink.

Log in to CloudLink Center to view detailed health status information.

Managing CloudLink syslog data

You can configure and view syslog data, and change the syslog message format. You can set security events that are visible in CloudLink Center to be automatically exported to a syslog server. Dell Technologies recommends that you configure CloudLink to forward this information through syslog to a centralized log server or a SIEM system.

Add Multiple Syslog Server Configuration

In Cloudlink 8.1 and higher, users can add up to four syslog servers.

About this task

Cloudlink 8.1 extends syslog functionality to include events, security events, audit logs, and authentication logs across multiple servers, enhancing monitoring capabilities beyond previous versions. Global customization of syslog format and the ability to postpone and resume actions are now available across multiple syslog servers. Syslog server deletion are performed individually, one at a time.

Steps

1. Log in to CloudLink Center.

2. Click **Server** > **Syslog** > **Add Syslog Server**.
3. In the **Add Syslog Server** dialog box, enter the following information:
   - From the **Server Status**—select any one of the available statuses:
     - Postponed—Logs are not sent out until the syslog status is explicitly set to Resume.
     - Active—Messages are sent to the configured syslog server until the Postpone command is selected.
   - In the **Host** box, enter the IP address of the host to store the system logs.
   - In the **Port** box, enter a port number to gather log messages.
   - From the **Protocol** list, select one of the following protocols:
     - **UDP**
     - **TCP**
     - **Secure TCP (TLS 1.2)**
   - From the **Facility** list, select a facility to log the syslog messages.
   - From the **Format** list, select one of the following message formats:
     - **CEF** (default)—The Common Event Format (CEF) uses the CEF:0|Device Vendor|Device Product|Device Version|Signature ID|Name|Severity|Extension message format.
     - **LEEF1**—The Log Event Extended Format (LEEF) version 1.0 uses the LEEF:1.0|Vendor|Product|Version|EventID|Extension message format.
     - **LEEF2**—LEEF version 2.0 uses the LEEF:2.0|Vendor|Product|Version|EventID|^|Extension message format.
     - **Custom**—The custom format is defined by an input string and variables.
4. Click **Add**.

View syslog configuration

Use the following procedure to view the Syslog status and configuration.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog**.
   The following information is displayed:
   - **Status**—Following are the available statuses:
     - Postponed—Logs are not sent out until the syslog status is explicitly set to Resume.
     - Active—Messages are sent to the configured syslog server until the Postpone command is selected.
   - **Host**—The host where system logs are sent.
   - **Facility**—The facility on which you log the syslog messages are logged.

Change syslog server configuration

The following procedure explains how to change the syslog server configuration.

About this task

If your organization requires the long-term retention of system events, it is recommended that you configure CloudLink to forward its security events to a centralized log server or SIEM using syslog. By default, CloudLink Center writes events to syslog using the common event format (CEF).

You can configure a syslog logger to direct all system log messages to the configured remote host. For information about network ports required, see the *Dell CloudLink Deployment Guide*.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog** > **Change Configuration**.
3. In the **Change Syslog Server Configuration** dialog box, enter the following information:
   a. In the **Host** box, enter the IP address of the host to store the system logs.
   b. In the **Port** box, enter a port number to gather log messages.
   c. From the **Facility** list, select a facility to log the syslog messages.

d. From the **Protocol** list, select one of the following protocols:
   - **UDP**
   - **TCP**
   - **Secure TCP (TLS 1.2)**

e. If you select **Secure TCP (TLS 1.2)** protocol, do the following:

   i. In the **Server Certificate** box, click  to select the server certificate.
   ii. In the **Certificate Format** list, select PEM certificate, or a password-protected PKCS12 (RC2) certificate.
   iii. (Optional) In the **PKCS12 Password** box, enter a password for the PKCS12 (RC2) certificate.
   iv. In the **Confirm PKCS12 Password** box, confirm the password for the PKCS12 (RC2) certificate.

4. Click **Change**.

Change syslog message format

Use the following procedure to change the syslog message format.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog** > **Change Syslog Format**.
   The **Change Syslog Message Format** dialog box is displayed.
3. From the **Syslog Format** list, select one of the following message format:
   - **CEF** (default)—The Common Event Format (CEF) uses the CEF:0|Device Vendor|Device Product|Device Version|Signature ID|Name|Severity|Extension message format.
   - **LEEF1**—The Log Event Extended Format (LEEF) version 1.0 uses the LEEF:1.0|Vendor|Product|Version|EventID|Extension message format.
   - **LEEF2**—LEEF version 2.0 uses the LEEF:2.0|Vendor|Product|Version|EventID|^|Extension message format.
   - **Custom**—The custom format is defined by an input string and variables.
4. Click **Change**.

8

Managing storage devices with CloudLink

You can use CloudLink to manage storage devices such as SSDs, NVMe drives, and self-encrypting drives (SEDs).

Encrypt SSDs or NVMe drives

CAUTION: Do not encrypt or erase drives that are added to a storage node. Only encrypt drives before they are added to a storage node, or after they are removed from a storage node. Any data on a drive is destroyed as a result of the following encryption procedures.

You can encrypt SSDs or NVMe drives using one of the following methods:

Encrypt devices in CloudLink Center

About this task

You can select the machine and drive to encrypt in CloudLink Center.

Steps

1. Click **Agent** > **Machines**.
2. Select the checkbox for the required machine.
3. Click **Actions** > **Encrypt**.
4. Select the checkbox next to the drive you want to encrypt.
5. Click **Encrypt**.
6. Repeat for each SSD or NVMe drive that requires encryption.

Encrypt a single drive with the CLI

About this task

You can connect to a storage node through SSH and use the CLI to run encryption commands.

Steps

1. Connect to the storage node using SSH.
2. To encrypt the drive, run:

   ```
   svm encrypt <device_name>
   ```

   where *<device_name>* is a variable.

   For example:

   ```
   svm encrypt /dev/sdX
   ```

   or

   ```
   svm encrypt /dev/nvmeXn0
   ```

   It takes about one minute to encrypt a new drive.

Encrypt multiple drives with the CLI

About this task

Use the following procedure to encrypt multiple drives with the CLI.

Steps

1. Connect to the storage node using SSH.
2. Encrypt the drives.
   - SAS-based SSDs:
     a. Select the drives for encryption.

        For example, `/dev/sdX`, where *X* could be `a` through `z`.

     b. Run the following command:

        ```
        for i in b c {j..o} {r..v};do svm -y encrypt /dev/sd$i;done
        ```
   - NVMe-based SSDs:
     a. Select the drives for encryption.

        For example, `/dev/nvmeXn1`, where *X* could be `0` through `24`.

     b. Run the following command:

        ```
        for i in {0..3} 6 7 {10..13} 16 17;do svm -y encrypt /dev/nvme$[i]n1;done
        ```

   It takes about one minute to encrypt a new drive.

Verify SSD or NVMe encryption

Find the new drive path of the SSD or NVMe and confirm that the drive is encrypted.

Steps

1. To find the new drive path, run:

   ```
   svm status
   ```

   - For SSDs, you will see output similar to the following: `/dev/mapper/svm_sdh`
   - For NVMe drives, you will see output similar to the following: `/dev/mapper/svm_nvme6n1`
2. In CloudLink Center, click **Agents** > **Machines**.
3. In the **Devices** area, verify that the drive is visible and has a status of **Encrypted**.

Manage self-encrypting drives

Manage the self-encrypting drives (SEDs) in CloudLink Center or using the CLI.

Manage self-encrypting drives in CloudLink Center

Manage the SED from CloudLink Center.

About this task

CAUTION: Never attempt to change management of SEDs that are attached to a storage node. Drives should only be managed before they are added to the storage node, or after removing them from the storage node. Any existing data on the drive is destroyed as a result of these procedures.

Steps

1. Click **Agent** > **Machines**.
2. Select the checkbox next to the SED that you want to manage.
3. Click **Actions** > **Manage SED**.
   The **Manage SED** dialog box is displayed. It is populated with the first unmanaged SED. You can add other SEDs to manage.
4. Click **Manage**.
5. Repeat for each SED that requires management.

Manage a single self-encrypting drive using the CLI

Use the CloudLink CLI to manage a self-encrypting drive (SED).

Steps

1. Connect to the storage node using SSH.
2. Run the following command to discover the unmanaged SEDs:

   ```
   svm status
   ```

3. Run the following command so that CloudLink manages the SED:

   ```
   svm manage /dev/sdX
   ```

   For example:

   ```
   svm manage /dev/sdh
   ```

4. Run `svm status` again to verify that the SED is now managed:

   NOTE: The SED status is displayed in the output as managed, but unencrypted.

   The output is similar to the following:

```
[root@rr5-o25-r650-node-03 ~]# svm status
State: Connected (server 100.65.23.225)
Group: Default
Policy: Manual
AES-NI HW acceleration: Yes

Volumes:
/                   unencrypted
swap                unencrypted

Devices:
/dev/sdf            managed             (sed    SZ: 1788G MOD: KPM6WRUG1T92     SPT: Yes
)
/dev/sdd            managed             (sed    SZ: 1788G MOD: KPM6WRUG1T92     SPT: Yes
)
/dev/sdb            managed             (sed    SZ: 3577G MOD: KPM6WVUG3T84     SPT: Yes
)
/dev/sde            managed             (sed    SZ: 1788G MOD: KPM6WRUG1T92     SPT: Yes
)
/dev/sdc            managed             (sed    SZ: 3577G MOD: KPM6WVUG3T84     SPT: Yes
)
/dev/sdb            unencrypted       (storagenode   SN:94917674              )
/dev/sdc            unencrypted       (storagenode   SN:94917675              )
/dev/sdd            unencrypted       (storagenode   SN:94917676              )
/dev/sde            unencrypted       (storagenode   SN:94917677              )
/dev/sdf            unencrypted       (storagenode   SN:94917678              )
/dev/sdg            encrypted         (storagenode   SN:94917679            /dev/mapper/
svm_sdg)
/dev/sdh            encrypted         (storagenode   SN:94917680            /dev/mapper/
svm_sdh)
/dev/sdi            encrypted         (storagenode   SN:94917681            /dev/mapper/
svm_sdi)
/dev/sdj            encrypted         (storagenode   SN:94917682            /dev/mapper/
```

```
svm_sdj)
/dev/sdk              encrypted        (storagenode    SN:94917683              /dev/mapper/
svm_sdk)
```

Verify self-encrypting drive encryption

Find the new device path and confirm that the self-encrypting drive (SED) is encrypted.

Steps

1. In CloudLink Center, click **Agents** > **Machines**.
2. Select the host of the encrypted drive.
3. Use the drive name listed in the `svm status` command output to perform the following verification checks:
   - In the **Devices** area, verify that the drive is visible and has a status of **Encrypted HW**.
   - In the **SED** area, verify that the status of the drive is displayed as **Managed**.

Add the encrypted drives to a storage node

After the drives are encrypted, use PowerFlex Manager to add them to the relevant PowerFlex storage node.

For software-encrypted SSDs, use the new `/dev/mapper/..` path when adding the drive. Self-encrypting drives (SEDs) use the normal `/dev/sdX` path.

CAUTION: Do not add a software-encrypted SSD using its original path, such as `/dev/`…. Doing so can stop the drive encryption, and this can corrupt the data on the drive.

Remove and erase drives

Use these procedures to remove a drive from a storage node and erase the drive. You can remove and erase both software-encrypted drives and self-encrypting drives (SEDs).

About this task

CAUTION: Never attempt to erase drives that are attached to a storage node. Only erase drives before they are attached to a storage node or after they are removed from a storage node. Erasing a drive destroys all data on the drive.

Steps

Before you remove a drive, confirm that the data on the drive is no longer required.

Use the web UI to remove a drive

About this task

Remove a drive from a storage node using PowerFlex Manager.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Lifecycle** > **Storage Nodes**.
3. From the list of storage nodes, select the checkbox beside the storage node that you want to remove a drive from and click **More Actions** > **Remove Device**.
4. In the **Remove Device** dialog box, verify that you selected the wanted drive and that the removal impacts the correct device group.
5. Type **`REMOVE DEVICE`** in the box and click **Remove**.

6. Verify that the operation is finished and is successful, and click **Dismiss**.

Use the web UI to erase a drive

About this task

Erase the drive using CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the checkbox next to the machine whose drive you want to erase.
4. Click **Actions** > **Erase**.
5. In the **Erase** dialog box, perform the following:
6. From the **Devices** list, select the required PowerFlex storage node drive.
7. In the **Confirm Device Name** box, enter the PowerFlex storage node drive name to confirm it, and then click **Erase**.

Use the CLI to remove and erase a drive

Steps

1. Run one of the following commands to remove a drive from a storage node using the PowerFlex CLI.
   - For SSDs, run the SCLI command `--remove_storagenode_device`. For example:

     ```
     scli --remove_storagenode_device --storagenode_name MY_STORAGENODE --device_path
     /dev/mapper/svm_sdX
     ```

   - For NVMe drives, run the SCLI command `--remove_storagenode_device`. For example:

     ```
     scli --remove_storagenode_device --storagenode_name MY_STORAGENODE --device_path
     /dev/mapper/svm_nvmeXn1
     ```

2. Run one of the following commands to erase the drive using the CloudLink CLI.
   - For SSDs, run:

     ```
     svm erase /dev/sdX
     ```

   - For NVMe drives, run:
   - ```
     svm erase /dev/nvmeXn0
     ```

Release management of a self-encrypting drive

You can use CloudLink Center to release CloudLink management of a self-encrypting drive (SED).

About this task

The **Release SED** option is only available if an SED is detected in the physical machine that is managed by CloudLink Center.

When CloudLink releases management of an SED, the SED is unlocked and CloudLink Center releases the encryption key.

NOTE: The **Release SED** option does not change any data on an SED. It only releases ownership of the encryption key.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the checkbox next to the machine whose SED you want to release.
     ```

4. Click **Actions** > **Release SED**.
5. From the **SEDs** list, select the required SED.
6. Click **Release**.

Results

The status of the SED drive changes to **Releasing Control**.

Once CloudLink releases management, the SED device status shows as **Unmanaged**.

NOTE: The **Release SED** option does not change any data on the SED.

Release management of a self-encrypting drive using the CLI

You can use the CLI to release CloudLink management of a self-encrypting drive (SED).

Steps

1. Log in to the storage node.
2. Run the following command:

```
svm release [device name]
```

For example:

```
svm release /dev/sdb
```

9

Modifying DNS entries

This section contains procedures on how to change the DNS entries. If you have multiple DNS servers that are configured but want to use a specific server, you can change the DNS entries to point only to that preferred server instead of listing all available servers.

Modifying DNS entries in the management virtual machines

Modify or update the DNS entries in the management virtual machines.

Steps

1. Establish an SSH connection to each of the management virtual machines nodes.
2. To view the existing DNS entries, run:

   ```
   sudo cat /etc/sysconfig/network/config | grep -i NETCONFIG_DNS_STATIC
   ```

3. To add or update the DNS entries:
   a. Run the following command:

      ```
      sudo vi /etc/sysconfig/network/config
      ```

   b. Add or modify the following lines:
      - **NETCONFIG_DNS_STATIC_SEARCHLIST=*example.com***, where *example.com* is the domain name
      - **NETCONFIG_DNS_STATIC_SERVERS=*x.x.x.x y.y.y.y***, where *x.x.x.x y.y.y.y* are the DNS server IP addresses.

   c. Press **Esc** and type **:wq** to save the file.
   d. To refresh and update the configuration, run:

      ```
      sudo netconfig update -f
      ```

4. Update the **coredns** pods on all the management virtual machines after making the DNS entry change for the pods in internal resolutions. Run the following command to view the pod names:

   ```
   kubectl get pods -n kube-system | grep rke2-coredns-rke2-coredns
   ```

5. To restart the **coredns** pods, delete them by running the following command:

   ```
   kubectl delete pods <pod-name> -n kube-system
   ```

   The pods are recreated after they are deleted.

Modifying the DNS entries in VMware vCenter

Modify the VMware vCenter DNS using VAMI.

Steps

1. Log in to VMware vCenter VAMI: **https://vcenter_ip_or_FQFN:5480**.
2. Go to the **Networking** section and in **Network Settings**, click **Edit**.

3. From **Edit Network Settings**, select **Network Adapter** and click **Next**.
4. In **Edit Settings**, expand the **Hostname and DNS** field, make the changes, and click **Next**.
5. In **SSO credentials**, enter the administrator credentials, for example, **`administrator@vsphere.local`**.
6. In **Ready to Complete**, review the changes, accept the acknowledgment, and click **Finish**.
7. A window appears with the message **Network update in progress**. When complete, the window displays **Network update succeeded**, click **Close**.

Modifying the DNS entries in the VMware ESXi host

Steps

1. Log in to the VMware ESXi host: **`https://ESXi_host_ip`**.
2. Click **Networking** > **TCP/IP Stacks**. Select **Default TCP/IP stack** and click **Edit Settings**.
3. Under **Basic DNS Configuration**, select **Manually configure the settings for this TCP/IP stack**.
4. Add or update the DNS IP addresses in the **Primary DNS server** and **Secondary DNS server** fields. In the **Search domain** field, add the domain name and click **Save**.

Modifying the DNS entries in the embedded Linux operating system

Modify or update the DNS entries in the embedded Linux operating system.

Steps

1. SSH to the embedded Linux operating system running on the storage node.
2. Go to the file path: `sudo vi /etc/sysconfig/network/config`
3. Add or modify the line:
   - **`NETCONFIG_DNS_STATIC_SEARCHLIST=`*`xxx.com`***, where *xxx.com* is the domain name
   - **`NETCONFIG_DNS_STATIC_SERVERS=`*`x.x.x.x y.y.y.y`***, where *x.x.x.x y.y.y.y* is the DNS server IP address
4. Press **Esc** and type **`:wq`** to save the file.
5. Type **`sudo netconfig update -f`** to refresh and update the configuration.

10

Backing up and restoring

Use the following procedures to back up and restore PowerFlex Manager, PowerFlex components, and CloudLink Center. Backups are critically important because they enable you to restore to a previously known state.

Back up and restore

Creating a backup saves all user-created data to a remote share from which it can be restored. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Perform frequent backups to guard against data loss and corruption. The best practice is to take a snapshot of PowerFlex Manager every time you perform a restore.

The **Backup** page displays information about the last backup operation that was performed on PowerFlex Manager. The information provided applies to both manual and automatically scheduled backups and includes the following:

- Last backup date
- Last backup status
- Back up directory path to a CIFS share
- Back up directory username

The **Backup** page also displays information about the status of automatically scheduled backups (enabled or disabled).

On this page, you can:

- Manually start an immediate backup
- Edit general backup settings
- Edit automatically scheduled backup settings

After performing a backup operation, you need to run a script outside of PowerFlex Manager to restore from the backup. The user interface does not support the ability to restore from a backup.

Editing backup settings and details

You can change the location where backup files are saved or the password that is required to access a backup file.

Steps

1. On the menu bar, click **Settings** and then click **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Settings**.
4. Indicate the network share location where the backup file is saved, enter a backup directory path in the **Backup Directory Path** box.

   Use the following format:

   - CIFS—`\\host\share`
5. If the username and password are required to access the network share, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes.
6. Click **Test Connection** to confirm that the backup settings you provided are correct.
7. To open the backup file, enter a password in the **Encryption Password** box.
8. To verify the encryption password, enter the password in the **Confirm Encryption Password** box.
9. To schedule automatic backups, next to **Scheduled Backups**, select **Enabled**. To discontinue automatically scheduled backups, deselect **Enabled**.
10. To specify the days on which backup must occur, select the days under **Frequency**.

11. From the **Run Time** drop-down list, select the time.
12. Click **Save**.

Backing up

In addition to automatically scheduled backups, you can manually run an immediate backup.

Steps

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Now**.
4. Select one of the following options:
   - To use the general settings that are applied to all backup files, select **Use Backup Directory Path and Encryption Password from Settings and Details**.
   - To use custom settings:
     a. In the **Backup Directory Path** box, enter a file path where the backup file is saved. Use this format:
        - CIFS—`\\host\share`

     b. Optionally, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes, if they are required to access the location you provided.
     c. Click **Test Connection** to confirm that the backup settings you provided are correct.
     d. In the **Encryption Password** box, enter a password that is required to open the backup file, and verify the encryption password by entering the password in the **Confirm Encryption Password** box.
5. Click **Backup Now**.

Restoring PowerFlex Manager

Restoring PowerFlex Manager returns user-created data to an earlier configuration that is saved in a backup file. To restore from a backup, you must run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Prerequisites

Before you begin the restore procedure, you must satisfy these prerequisites:

- The restore cluster must be the same PowerFlex version and Kubernetes version.
- The restore cluster must have the same IP addresses and configuration.

  The cluster configuration must be the same as the cluster configuration where the backup was taken.

  - All Kubernetes nodes must have the same IP addresses.
  - All Kubernetes nodes must have the same names.
  - All LoadBalancer IP addresses must be the same.

About this task

CAUTION: Restoring an earlier configuration restarts PowerFlex Manager and deletes data that is created after the backup file to which you are restoring. Any running jobs could be terminated as well.

Steps

1. Log in, using the username and password used for the installer VM, to the node where the PowerFlex Manager platform installer was initially run.
2. Run the restore script that is included with the installer bundle:

   ```
   ./restore_backup.sh
   ```
3. Provide details to complete the execution of the script.

You are prompted to provide the SSH username and password. In addition, you are asked to specify whether the passwords are the same for all nodes, and also provide the location of the backup file and the encrypted password for the file. You may also be asked to provide the CIFS username and password. The CIFS credentials may not be required for a CIFS share that allows for anonymous access.

NOTE: The passwords must be in base64 encoded format.

To complete the execution of the restore script, you must specify whether the restore operation will be performed on an existing cluster or a new cluster.

Here is a snippet that shows a sample that is run of the restore script:

```
[root@sheetal-installer scripts]# ./restore_backup.sh

/usr/local/lib/python3.8/site-packages/paramiko/transport.py:236:
CryptographyDeprecationWarning: Blowfish has been deprecated "class":
algorithms.Blowfish, Installation logs are available at <Bundle root>/PFMP_Installer/
logs/ directory. More detailed logs are available at <Bundle root>/atlantic/logs/
directory. PFMP Installer is about to reset a PFMP cluster based on the configuration
specified in the PFMP_Config.json.

Please enter the ssh username for the nodes specified in the
PFMP_Config.json[root]:root

Are passwords the same for all the cluster nodes[Y]?:Y

Please enter the ssh password for the nodes specified in the PFMP_Config.json.
Password:

Please enter backup file location - Example CIFS: \\1.2.3.4\Lab\backup\file.tgz

Please enter CIFS username. Press enter to skip if username is not
required:administrator

Please enter CIFS password(base64 encoded). Press enter to skip if username is not
required: UmFpZDR1cyE=

Please enter encryption password for backup file (base64 encoded): UmFpZDR1cyE=

Perform restore on existing cluster? Please enter yes/y or no/n :
```

Results

The restore process prints status information until the restore is complete.

Related tasks

Restoring compliance file data

Restoring compliance file data

Use this procedure to return compliance files to a healthy state.

About this task

After performing a restore operation, the compliance files will be in an ERROR state. The compliance file data is not part of the backup payload, so after a restore, the compliance files appears in an error/unhealthy state. Any changes previously made to compliance files using single component upgrade, for example BIOS, iDRAC, or ESXi version modifications, will not be preserved through the backup and restore process.

Steps

1. Resynchronize the compliance file from the original source to restore it to a good state.
2. If any components were previously modified using single component upgrade (BIOS, iDRAC, ESXi depot), edit the compliance file using single component upgrade to reapply those changes.
3. Confirm that the compliance file is no longer in ERROR state and all components reflect the expected versions.

Related tasks

Restoring PowerFlex Manager

Integrated Dell Remote Access Controller 9

You can back up and restore integrated Dell Remote Access Controller (iDRAC) configuration settings.

Back up iDRAC

You can back up the system and configuration settings for iDRAC. The backup creates a file that you can save locally or on a network share (CIFS, NFS, HTTP, or HTTPS).

Steps

1. Log in to the iDRAC using root credentials.
2. Click **Configuration** > **Server Configuration Profile**.
3. Click **Export**.
4. Select one of the following to specify the location type:

   CIFS, NFS, HTTP, or HTTPS

   - **Local** to save the configuration file on a local drive.
   - **Network Share** to save the configuration file on a CIFS or NFS share.
   - **HTTP or HTTPS** to save the configuration file to a local file using HTTP/HTTPS file transfer.

   Depending on the location type, you must enter the Network Settings or HTTP/HTTPS settings. If the proxy is configured for HTTP/HTTPS, proxy settings are also required.
5. Select the components whose configuration you must back up.
6. Select the **Export** type:
   - **Basic**
   - **Replacement Export**
   - **Clone Export**
7. Select an **Export file format**.
8. Select **Additional export items**.

Restore iDRAC

You can restore iDRAC from a backup when you want to use previous configuration settings.

Steps

1. Click **Configuration** > **Server Configuration Profile**.
2. Select one of the following to specify the location type:
   - **Local** to save the configuration file on a local drive.
   - **Network Share** to save the configuration file on a CIFS or NFS share.
   - **HTTP or HTTPS** to save the configuration file to a local file using HTTP/HTTPS file transfer.

   Depending on the location type, you must enter the Network Settings or HTTP/HTTPS settings. If the proxy is configured for HTTP/HTTPS, proxy settings are also required.
3. Select the components listed in the **Imported Components** option.
4. Select the **Shutdown** type.
5. Select the maximum wait time to specify the wait time before the system shuts down after the import is complete.
6. Click **Import**.

Network switch configuration back up and restore

Back up and restore running configuration of the Cisco and Dell switches.

About this task

If you own the switches, take the switch backup immediately after successful PowerFlex deployment.

NOTE: You are responsible for providing the backup location and maintaining the backup as well. Dell Technologies recommends storing the backup in a separate shared location, for example, not on the jump server.

Steps

1. Connect to the Cisco Nexus or Dell PowerSwitch switch, either using the console cable, Telnet, or SSH using admin credentials, type:

   ```
   #copy running-config scheme://server/[url/]filename
   ```

   For the scheme argument, you can enter **tftp:**,**ftp:**, **scp:**, or **sftp:**.

   The server argument is the address or name of the remote server, and the URL argument is the path to the source file on the remote server. The server, URL, and file name arguments are case-sensitive.

   For example:

   ```
   switch# copy running-config tftp://10.10.10.1/sw1-run-config.bak

   switch# copy running-configuration scp://root:calvin@10.11.10.12/tmp/backup.txt
   ```

2. Restore the network configuration, connect to the Cisco Nexus or Dell PowerSwitch switch, either using the console cable, Telnet, or SSH using admin credentials, type:

   ```
   #copy scheme://server/[url/]filename running- config
   ```

   For the scheme argument, you can enter **tftp:**,**ftp:**, **scp:**, or **sftp:**.

   The server argument is the address or name of the remote server, and the URL argument is the path to the source file on the remote server. The server, URL, and file name arguments are case-sensitive.

   For example:

   ```
   switch# copy tftp://10.10.10.1/my-config running-config
   switch# copy scp://root:calvin@10.11.10.12/tmp/backup.txt running-configuration
   ```

Backing up and restoring CloudLink Center

Use this procedure to create a CloudLink Center backup file.

View backup information

Viewing backup information helps ensure data recovery, supports compliance, aids troubleshooting, enables version control, and facilitates system upgrades or migrations.

To view the backup information, log in to the CloudLink Center, and click **System** > **Backup**. The **Backup** page lists the following information:

- **Backup File Prefix**—Prefix used for the backup files.
- **Current Key ID**—The identifier for the current RSA-2048 key pair.
- **Current Backup File**—The name of the current backup file.
- **Current Backup Time**—The date and time that the current backup file was generated.
- **Backup Schedule**—The schedule for generating automatic backups.
- **Next Backup In**—The time remaining before the next automatic backup is generated.

When a backup file is downloaded, the **Backup** page lists the following additional information:

- **Last Downloaded File** —The name of the backup file that was last downloaded. Only shown when a backup file has been downloaded.
- **Last Downloaded Time** —The date and time of the last backup file download. Only shown when a backup file has been downloaded.
- **Backup Store** —The backup store configuration type. If you have not configured a backup store, the value is Local, which is stored on the local desktop.

  You can also use the FTP or SFTP servers as backup stores. To change the backup store, click **Actions** > **Change Backup Store**.

  If you have configured an FTP or SFTP backup store, the following additional information is available:

  - **Host** —The remote FTP, SFTP, or FTPS host where you saved the CloudLink Center backups. You can set this value to the host IP address or hostname (if DNS is configured).
  - **Port**—The port used to access the backup store.
  - **User**—The user with permission to access the backup store.
  - **Directory**—The directory in the backup store where backup files are available.

Change the schedule for automatic backups

CloudLink Center automatically generates a backup file each day at midnight (UTC time).

To change the schedule for generating automatic backups, click **System** > **Backup** > **Actions** > **Change Backup Schedule**.

Generate a backup file manually

Use this procedure to generate a backup manually, if you want to preserve CloudLink Center before the next automatic back up. Download the backup file when you generate a backup file manually.

Steps

1. Log in to the CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Generate New Backup**. A backup file is generated.

Generate a backup key pair

You can generate a new backup key pair. For example, if the private key for a backup key pair is lost, you can generate a new key pair. You cannot access your backup files without the associated private key. When you generate a new key pair, CloudLink Center automatically generates a new backup file to ensure that the current backup can be opened with the private key of the current key pair.

Dell Technologies recommends the following practices when you generate a new backup key pair.

1. Download the private key to the Downloads folder for the current user account. For example, `C:\Users\Administrator\Downloads`.

   NOTE: The previously generated backup key will not open the backup file created, after a new key is generated.

2. Generate the new key pair by clicking **System** > **Backup** > **Actions** > **Generate and Download New Key**.
3. Click **Generate and Download**.

Download the current backup file

You can download the current backup file at any time. The current backup file is either:

- The last backup file that CloudLink Center automatically created.
- The last backup file that you manually generated after the last automatic backup.

To download the backup file:

1. Click **System** > **Backup** > **Actions** > **Download Backup**.
2. In the **Download Current Backup** dialog box, click **Download**.

When you download the current backup file, CloudLink Center shows the age of the backup file.

Restore the CloudLink backup

Use this procedure to restore the CloudLink backup.

Steps

1. Log in to the CloudLink Center.
2. Click **System** > **Backup** > **Actions** > **Restore Keystores**.
3. In the **Restore Keystores** dialog box, complete the following steps:
   a. In the **Key** box, browse to the private key file.
   b. In the **Backup** box, browse to the backup file.
   c. In the **Unlock** box, type the passcode that was set during the initial configuration of the CloudLink Center.
   d. Click **Restore**.
   A **Restore Keystores succeeded** message is displayed.

   NOTE: If the CloudLink backup is not associated with a key pair, the file is corrupted or key mismatch error message is displayed. In such a scenario, generate a new key pair and download the backup file.

11

Managing usernames and passwords

This section contains procedures to change iDRAC, VMware vSphere Client, CloudLink and system passwords, generate SSH key pairs, and manage credentials.

PowerFlex servers

Use the following procedures to change server iDRAC and serve BIOS passwords.

Change the Integrated Dell Remote Access Controller 9 (iDRAC9) password

Use the iDRAC web console to change the root password.

Steps

1. Open a web browser and type: **`http://<ip_address_of_iDRAC>`**
2. Log in as **`root`**.
3. Expand **iDRAC Settings**.
4. Click **Users** > **Local Users**.
5. Select **User ID 2** and click **Edit**.

   ID 2 is the root user.
6. Under **User Configuration**, in the **User Account Settings** section, change the password.
   a. In the **Password** box, enter the new password.
   b. In the **Confirm Password** box, re-enter the new password.
7. Click **Save**.

Change the system and setup passwords

Use this procedure to change the system BIOS password.

Prerequisites

If the password status is set to `locked`, you cannot change or delete the system or setup password.

Steps

1. Enter System Setup by pressing **F2** immediately after turning on or restarting your system.
2. On the System Setup Main Menu, click **System BIOS** > **System Security**.
3. On the **System Security** screen, ensure that **Password Status** is set to **`Unlocked`**.
4. In the **System Password** field, change or delete the existing system password and press **Enter**.
5. In the **Setup Password** field, change or delete the existing setup password and press **Enter**.

   If you change the system or setup password, a message prompts you to re-enter the password. If you delete a password, a message prompts you to confirm the deletion.
6. Press **Esc** to return to the **System BIOS** screen. Press **Esc** again. A message prompts you to save the changes.

Generate SSH key pairs

To create the SSH keys for a PowerFlex node, switch, **OS Admin**, **OS User**, use this procedure to manually generate the SSH keys that are of RSA type and ensure the key is generated without a passphrase.

Steps

1. Generate a private key using the following command:

   ssh-keygen -b 2048 -t rsa

2. Enter a file path to save the public and private keys or proceed with the default file path without a passphrase:

   Sample output:

   ```
   Enter file in which to save the key (/root/.ssh/id_rsa):
   Enter passphrase: # do not enter a passphrase
   Enter same passphrase again: # do not enter a passphrase
   ```

   Sample output that indicates the public and private keys are saved in the default file path:

   ```
   Your identification has been saved in /root/.ssh/id_rsa
   Your public key has been saved in /root/.ssh/id_rsa.pub
   The key fingerprint is:
   SHA256:fKIllqp5UXeuEW7xuYeE7PNHpvjAUatNx6I/bisRqRU root@PFMP2-4
   The key's randomart image is:
   +---[RSA 2048]----+
   |                 |
   |        E        |
   |         o.      |
   |       .oB..o    |
   |      .+*SO=.o   |
   |     .o.=BO=oo   |
   |     ...o=*.*    |
   |    o.    *o* o  |
   |   o.      B*=   |
   +----[SHA256]-----+
   ```

3. Verify that the public and private keys are saved in the default file path: `/root/.ssh/`.

   Sample output:

   ```
   cd /root/.ssh
   ls
   ```

   - `id_rsa.pub` indicates a public key.
   - `id_rsa` indicates a private key.

Change the management VM password

Use the following procedure to change a user's password for the management VMs.

About this task

See *PowerFlex Manager default accounts* in the *Dell PowerFlex Family Security Configuration Guide* for the default PowerFlex management platform cluster VM account.

Steps

1. Log in to the appropriate user account. If you are changing a password for another user, log in as the root user or a user with root privileges.
2. Open a shell prompt and type one of the following commands:
   - To change your own password: **passwd**
   - To change the password of another user: **passwd username**

When prompted, enter and confirm the new password.

Changing the operating system password

If you lose the password on storage VM (SVM), start up in recovery mode to reset it.

Steps

1. Log in to PowerFlex Manager.
2. Click **Lifecycle** > **Storage Nodes**.
3. Select **Node** > **More actions** > **Enter Maintenance Mode**.
4. Log in to VMware vCenter. Select the cluster and click **Related Objects** > **Virtual Machines** > **Open Console**.
5. Restart the SVM.
6. Enter **ESC** during restart and click **OK** to enter text mode.

   NOTE: You might have to hold down the **ESC** key.

7. Type `E` to edit the password.
8. Select the second line (starts with `kernel /boot…`) and type `E`.
9. Append `init=/bin/bash` to the end of the line.
10. Select `B` to restart in recovery mode.
11. After the PowerFlex node restarts, type `passwd`, and enter a new password.
12. Type `reboot`, and allow the SVM to restart using the new password.

Change CloudLink passwords

CloudLink software is installed on the CloudLink Center VMs.

Unlock secadmin user password

Use this procedure to unlock the secadmin password.

Steps

1. Log in to the controller vCenter web client, and launch the CloudLink Center VM console.

   NOTE: For a single vCenter environment, log in to single vCenter with a controller and customer datacenter. Go to the controller datacenter to launch the CloudLink center VM console.

2. Log in using the CloudLink user credentials.
3. In the **Update Menu** dialog box, select **Unlock User**, and click **OK**.
   A user unlocked message is displayed.
4. Click **OK**.

VMware virtualization credentials

Use the following procedures to change credentials for VMware ESXi, VMware vSphere Client, VMware vCenter Server, and VMware vCenter Server Appliance.

Changing a VMware ESXi host root password

For security reasons it might be necessary to change the password for the root user on a VMware ESXi host after installation.

Use any of the following methods to change the root password for the VMware ESXi host:

- VMware vSphere Client
- VMware ESXi shell command
- VMware ESXi host System Customization menu

Change the password using VMware vSphere Client

Steps

1. Log in to the host UI https://esxi_IP_address using root credentials.
2. In the left pane, under **Host**, click **Manage**.
3. Select the **Security & Users** tab, click **Users** and click the **root** user.
4. Click **Edit User**.
5. In the **Edit User** dialog box, enter and confirm a new password.
6. Click **Save**.

Change the password using the VMware ESXi shell command

Prerequisites

Log in to the VMware ESXi host service console as the root user.

Steps

1. When prompted, enter the current password.
2. To change the root password, enter: **passwd root**.
3. Enter the new root password and press **Enter**.
4. Verify the password by entering it again.

Change the password using the VMware ESXi host system customization menu

Prerequisites

Log in to the VMware ESXi host service console as **root** user.

You can also acquire root privileges by executing the **su** command.

Steps

1. From the **System Customization** menu of the VMware ESXi host, use the keyboard arrows to select **Configure Password** and press **Enter**.
2. In the **Configure Password** dialog box, fill in the required fields to change the password:
   a. Enter the **Old Password** of the VMware ESXi host.
   b. Enter the new root password in the **New Password** field. Re-type it in the **Confirm Password** field.
   c. Press **Enter**.

Modify the VMware vCenter Server single sign on default administrator password

Use this procedure to change the default password for the VMware vCenter single sign on administrator account.

Prerequisites

Log in to the VMware vSphere Client and connect to vCenter.

Steps

1. In the left pane, select **Administration**.
2. Under Single Sign On, select **Users and Groups**.
   The **admin** user displays in the right pane.
3. On the **Users** tab, select the domain from the drop-down menu, and select the **administrator** user. Click **Edit**.
4. Set and confirm the password for the admin user account. Be sure to use a strong password as the system validates the password before accepting it.
5. Click **Save**.

Change the administrator password for VMware vCenter Server Appliance

Use this procedure to change the VMware vCenter Server Appliance administrator password.

Steps

1. Log in to the VMware vCenter Server Appliance Web console.
2. On the **Admin** tab, enter your current password in the **Current administrator password** box.
3. Enter the new password in the **New administrator password** and **Retype new administrator password** boxes.
4. Click **Change password**.

Reset the VMware vCenter Server Appliance root password

Use this procedure to reset a forgotten vCenter Server Appliance root password.

Steps

1. Take a snapshot or backup of the vCenter Server Appliance. Do not skip this step.
2. Reboot the vCenter Server Appliance.
3. After the operating system starts, press the **E** key to access the **GNU GRUB Edit Menu**.
4. Locate the line that starts with the word `Linux`. Append the following entries to the end of the line:

   ```
   rw init=/bin/bash
   ```

5. Press **F10**.
6. At the command prompt, type the following command: **`passwd`**
7. Enter a new root password and re-enter the password to confirm it.
8. Unmount the file system by running the following command: **`umount /`**
9. Reboot the vCenter Server Appliance by running the following command: **`reboot -f`**
10. Confirm that you can access the vCenter Server Appliance using the new root password.
11. Remove the snapshot taken in step 1 if applicable.

12

Retrieving logs

You can retrieve logs and support data for operating systems, PowerFlex components, the PowerFlex management platform, the PowerFlex installer, RAID controllers, and system events. You can also generate a comprehensive troubleshooting bundle.

Retrieving operating system logs

You can retrieve logs from a VMware ESXI-based operating system, a Linux-based operating system, or a Windows-based operating system. You can also configure and retrieve operating system crash dumps.

Related information

Retrieve logs from a VMware ESXi-based operating system
Retrieve logs from a Linux-based operating system
Retrieve logs from Windows-based operating system
Configure and retrieve operating system crash dumps

Retrieve logs from a VMware ESXi-based operating system

Use this task to retrieve the logs from a VMware ESXi-based PowerFlex node.

Prerequisites

Ensure that you have access to the following:

- IP address of the node
- Password for the root user

About this task

When the operating system is started, the ESXi operating system logs are collected in the system-defined folders `/var/log` and `/scratch/log/` on the local drive.

Steps

1. Log in to the PowerFlex node.
2. From ESXi, run the `scp` command to collect the ESXi operating system logs from the following folders on the local drive:
   - `/var/log`
   - `/scratch/log/`

   For additional data, see KB article: Collecting diagnostic information for VMware ESXi using vSphere Client

Next steps

An inventory job is scheduled to run every day at [HH:mm] to identify the resources that are running in your system. If an upgrade operation is in progress simultaneously, the upgrade operation overrides the job and the job is skipped.

If you turn off the inventory job setting, the job is paused for 24 hours. After 24 hours, the inventory job setting automatically turns back on and the inventory job continues running from the time it was paused.

Related information

Retrieving operating system logs

Retrieve logs from a Linux-based operating system

Use this task to retrieve the logs from a Linux-based PowerFlex node.

Prerequisites

Ensure that you have access to the following:

- IP address of the node
- Password for the root user
- Grab for Unix/Linux, available at Dell Technologies Central.

About this task

When the Linux server is started, the Linux logs are collected in the system-defined directory `/var/log`, on the local drive.

Operating system logs are also collected automatically when running the `get_info.sh` core component log collection script. For information on running `get_info.sh`, see *Retrieve core component logs with get_info*.

Steps

1. Log in to the PowerFlex node.
2. Run the `scp` command to collect the Linux operating system logs from the `/var/log` directory on the local drive.

   The `/var/log` folder provides logs for preliminary system investigation.
3. To retrieve full logs, including customer- and OS-related information, run Grab for Unix/Linux.

Related information

Retrieving operating system logs

Retrieve logs from Windows-based operating system

Use this task to retrieve the logs from a Windows-based PowerFlex node.

Prerequisites

Ensure that you have access to the following:

- IP address of the node
- Password for the administrator user
- Grab for Windows, available at Dell Technologies Central.

Steps

1. Log in to the PowerFlex node.
2. Run the Windows Event Viewer.

   The **Event Viewer** screen displays **Event Viewer (Local)** as the root node.
3. In the navigation pane, go to the **Windows Logs** node and select the **System** sub node.

   The **Actions** pane is displayed on the right side of the screen.
4. From the **Actions** pane, click **Save All Events As**.

   The navigation tree with your local folders is displayed.
5. Browse to the intended location on your local drive, give the file an appropriate name, and select the file format.
6. Click **Save**.

   The Windows OS-related log files are saved on your local drive.

   The file on the local drive provides logs for preliminary system investigation.
7. To retrieve full logs, including customer and OS-related information, run Grab for Windows.

Related information

Retrieving operating system logs

Configure and retrieve operating system crash dumps

Use this task to configure and retrieve the operating system crash dumps.

Steps

1. Open a web browser.
2. To configure and retrieve the crash dumps, click the link that matches with your operating system:

Operating System	Link for crash dump configuration procedure
ESXi	Configuring an ESXi/ESX host to capture a VMkernel coredump
Linux (RedHat)	How to troubleshoot kernel crashes, hangs, or reboots with kdump on Red Hat Enterprise Linux
Linux (SLES)	Configure crashkernel memory for kernel core dump analysis
Windows	Generate a kernel or complete crash dump

3. To analyze the crash dumps, click the link that matches with your operating system:

Operating System	Link for crash dump configuration procedure
Linux (RedHat)	How can I use crash to send Red Hat some vmcore pre-analysis information before or while uploading the vmcore image?
Linux (SLES)	Generating a Kernel Core Dump Analysis File
Windows	How to read the small memory dump file that is created by Windows if a crash occurs

Related information

Retrieving operating system logs

Retrieving component logs

You can retrieve PowerFlex component logs using the `get_info` script or the PowerFlex REST API.

Related information

Retrieve core component logs with get_info
Retrieve PowerFlex core component logs using the API

Retrieve core component logs with get_info

The `get_info` script is a debug utility which allows a technician to collect debug information and send it back to Technical Support for further analysis. You can manually retrieve core component logs from one server node at a time.

Prerequisites

Ensure that you have the following:

- IP address of the node
- Password for the root user in Linux-based systems and the administrator user in Windows-based systems

About this task

The logs relate to the following core components on each server node and its operating system. Core components are the userland components and the kernel component. The following component lists are only applicable to PowerFlex systems that use the erasure coding protection scheme.

Userland components:

- `dgwt` (device gateway target)
- `lia` (lightweight installation agent)
- `mdm` (metadata manager)
- `pds` (PowerFlex data server)
- `sdt` (storage data target)

Kernel component:

- `sdc` (storage data client)

If the selected node contains more than one component, the script gathers logs for all components on that node.

The `get_info` syntax and arguments are explained in *Optional arguments for get_info*.

NOTE: Only one instance of `get_info` can run at a given time. If there is not enough space for its output, `get_info` does not run.

Steps

1. Use SSH or RDP to connect to the node.
2. Run the script for the component.

   Linux example:

   ```
   /opt/emc/scaleio/<PowerFlex_component>/diag/get_info.sh
   ```

   The *<PowerFlex_component>* value is dgwt, lia, mdm, pds, sdt, or sdc.

   Windows example:

   ```
   "C:\Program Files\EMC\scaleio\<PowerFlex_component>\diag\get_info.bat" -f
   ```

   The *<PowerFlex_component>* value is dgwt, lia, mdm, pds, sdt, or sdc.

   - If the selected node is the primary MDM, use the flags `-u <MDMuser> -p <MDMpassword>` instead of `-f`.
   - If the selected node contains more than one PowerFlex component, running any `get_info` script gathers logs for all components on that node.

   When the log collection process is complete, an archive file containing the logs of all components in the node is created in a temporary directory. The archive file is a TGZ file on Linux hosts and a ZIP file on Windows hosts.

   The default directory locations are `/tmp/scaleio-getinfo` on Linux hosts and `C:\Windows\Temp\ScaleIO-getinfo` on Windows hosts.
3. Verify that output similar to the following is returned, which shows that the process of log collection was completed successfully:

   ```
   bundle available at '/tmp/scaleio-getinfo/getInfoDump.tgz'
   ```

   NOTE: The script can generate numerous lines of output. It is recommended that you search for this line in the output.
4. Retrieve the log file.

Related information

Retrieving component logs
Optional arguments for get_info

Optional arguments for get_info

The `get_info` script uses several optional arguments. The script contains a list of all arguments, with descriptions and examples. The script uses the older PowerFlex product name, ScaleIO.

Syntax

```
get_info.sh [OPTIONS]
```

Optional arguments

-a, --all

Collect all data

-A, --analyse-diag-coll

Analyze diagnostic data collector (diag coll) data

-b[COMPONENTS], --collect-cores[=COMPONENTS]

Collect existing core dumps of the space-separated list of user-land components, COMPONENTS (default: all user-land components)

Example: `-b'mdm pds'` (no space between option name and COMPONENTS)

Example: `--collect-cores='mdm pds'` (separate option name and COMPONENTS with a single equal sign, "=")

-d OUT_DIR, --output-dir=OUT_DIR

Store collected bundle under directory `OUT_DIR` (default: *<WORK_DIR>*/scaleio-getinfo, see `--work-dir` for the value of *<WORK_DIR>*)

-f, --skip-mdm-login

Skip query of PowerFlex login credentials

-J, --xz

Use tar/xz format for the collected bundle

-k NUM, --max-cores=NUM

Collect up to NUM core files from each component (default: all core files)

Implies --collect-cores

-l, --light

Generate light bundle (not recommended)

--ldap-authentication

Log in to PowerFlex using LDAP-based authentication

-m NUM, --max-traces=NUM

Collect up to `NUM` PowerFlex trace file from each component (default: all files)

--management-system-ip=ADDRESS

Connect to SSO or management and orchestration at `ADDRESS` for PowerFlex log in (default: scli default)

--mdm-port=PORT

Connect to MDM using `PORT` for SCLI commands (default: scli default)

-n, --use-nonsecure-communication

Connect to the MDM in non-secure mode

-N, --skip-space-check

Skip free space verification

--overwrite-output-file

Overwrite output file if it already exists

-p PASSWORD, --password=PASSWORD

Use `PASSWORD` for PowerFlex log in (default: scli default)

-P PATH, --collect-path=PATH

Collect path `PATH`

This option can be used multiple times to collect multiple paths

--p12-password=PASSWORD

Encrypt PowerFlex login PKCS#12 file using `PASSWORD` (default: scli default)

--p12-path=FILE

Store PowerFlex login PKCS#12 file as `FILE` (default: scli default)

-q, --quiet, --silent

Do not output messages to stdout

-r, --mdm-repository

Collect MDM repository files

-s, --skip-sdbg

Skip collection of SDBG output

-S, --pause-core-generation

Pause core generation of PowerFlex components during data collection

--tech

Include technician option in help message

-u USERNAME, --username=USERNAME

Use `USERNAME` for PowerFlex log in (default: scli default)

-w WORK_DIR, --work-dir=WORK_DIR

Use directory `WORK_DIR` for temporary files (default: /tmp)

-x FILE, --output-file=FILE

Store collected bundle as `FILE` (default: getInfoDump). Appropriate file name suffix (.tgz, .zip) is added automatically

If FILE is '-', write bundle to standard output (implies --quiet)

-z, --zip

Use zip format for the collected bundle

-h, --help

Show this help message and exit

Related information

Retrieve core component logs with get_info

Retrieve PowerFlex core component logs using the API

You can use the PowerFlex REST API to collect core component logs.

Prerequisites

- The PowerFlex management platform (PFMP) cluster is operational. The PFMP was previously called Management and Orchestration (mno).
- The PowerFlex cluster is deployed.

- The PowerFlex management platform cluster can communicate with any host instance in the PowerFlex cluster. To test the connection, ping from a PowerFlex node to the data IP addresses of the instances.
- Ensure that you have the following information:
  - **mdm_ips**
  - **mno_username**
  - **mno_password**
  - **mno_ip**

Steps

1. To log in to the PowerFlex management platform, run the following API command:

   `POST https://<mno_ip>/rest/auth/login`

   For example:

   ```
   token=$(curl -k --silent 'https://<mno_ip>/rest/auth/login' --header 'Accept:
   application/json' --header 'Content-Type: application/json' -d '{"username":
   "<mno_username>","password": "<mno_password>"}' |jq -r '.access_token'); echo $token
   ```

   Save the access token received in the output.
2. To get the configuration, run the following API command:

   `POST https://<mno_ip>/im/types/Configuration/instances`

   For example:

   ```
   curl --silent  -k -X POST  -H "Authorization: Bearer ${token}" -H
   'Content-Type: application/json' -d '{ "mdmIps":["10.234.177.103","10.234.177.153"],
   "mdmUser":<mno_username>, "mdmPassword":"<mno_password>",
   "securityConfiguration":{ "allowNonSecureCommunicationWithMdm":"false",
   "allowNonSecureCommunicationWithLia":"false", "disableNonMgmtComponentsAuth":"false"
   } }'  https://<mno_ip>/im/types/Configuration/instances>config.json
   ```

   Save the output of the request. This output is a JSON representation of the system configuration.
3. Add the M&O login information to the JSON. The get info script requires the login information to perform certain queries, to provide this information, add the following key value pairs to JSON:

   ```
   mnoUser : "<mno_username>"
   mnoPassword : "<mno_password>"
   mnoIp : "<mno_ip>"
   ```

   For example:

   ```
   {
   "mnoUser" : "<mno_username>" ,
   "mnoPassword": "<mno_password>" ,
   "mnoIp: "<mno_ip>"
   "snmpIp": null
   … (rest of the json)
   }
   ```

4. To collect logs, run the following API command:

   `POST https://<mno_ip>/im/types/NodeInfo/instances/actions/collectLogs`

   To run the collect logs request, ensure that you have a valid token and the configuration JSON. For more optional attributes, go to step 7.

   For example:

   ```
   curl -k -X POST  -H "Authorization: Bearer ${token}" -H 'Content-Type:
   application/json' -d @config.json  https://<mno_ip> /im/types/NodeInfo/instances/
   actions/collectLogs
   ```

5. To monitor the log collection process, run the following API command:

   `GET https://<mno_ip>/im/types/ProcessPhase/actions/queryPhaseState`

Monitor the `phaseStatus` value and wait until the operation is complete.

For example:

```
curl -s -k  -X GET -H "Content-Type:application/json" -H "Authorization: Bearer $
{token}" https://<mno_ip> /im/types/ProcessPhase/actions/queryPhaseState
```

Output example:

```
{"phaseStatus":"completed","phase":"query","numberOfRunningCommands":0,"numberOfPendin
gCommands":0,"numberOfCompletedCommands":6,"numberOfAbortedCommands":0,"numberOfFailed
Commands":0,"failedCommand
```

6. To download the logs, run the following API command:

   `GET https://<mno_ip>/im/types/NodeInfo/instances`

   For example:

   ```
   curl -s -k -i -X  GET -H "Content-Type:application/json" -H "Authorization: Bearer
   ${token}" https://<mno_ip>/im/types/NodeInfo/instances -o "get_info.zip"
   ```

   In this example, the logs are downloaded to the current working directory in an archive file named `get_info.zip`.

7. (Optional) Add the following optional attributes to the log collection request as query parameters:
   - `targetIPs`—Filter the result to only part of the nodes (by their IP addresses)
   - `copyRepositories`—Copy MDM repositories (true/false, default is false)
   - `liteVersion`(lite version)—Collect trc.0, exp.0, and umt.0 files only and repository files (true/false, default is false)
   - `copyBinaries`—Collect MDM, DGWT, PDS, SDT, LIA binaries, and core dumps (true/false, default is false)
   - `collectSdbgScreens` (true/false, default is false)

   For example:

   ```
   curl  -k -X -i POST -H "Content-Type:application/json" -H "Authorization:
   Bearer ${token}"   "https://<mno_ip>/im/types/NodeInfo/instances/actions/collectLogs?
   copyRepositories=true&targetIPs=10.55.118.61&targetIPs=10.55.118.62"
   ```

8. To clear the operation and move to idle, run the following API commands:

   `POST https://<mno_ip>/im/types/Command/instances/actions/clear`

   `https://<mno_ip>/im/types/ProcessPhase/actions/moveToIdlePhase`

   After the log collection operation is complete, mark the operation as completed to allow the other jobs to run.

   For example:

   ```
   curl -s -k  -X POST -H "Content-Type:application/json" -d '{}' -H "Authorization:
   Bearer ${token}" https://<mno_ip>/im/types/Command/instances/actions/clear
   ```

   ```
   curl -s -k  -X POST -H "Content-Type:application/json" -d '{}' -H "Authorization:
   Bearer ${token}" https://<mno_ip>/im/types/ProcessPhase/actions/moveToIdlePhase
   ```

Related information

Retrieving component logs

Retrieving additional PowerFlex logs

You can retrieve additional logs for the PowerFlex management platform, RAID controllers, and system events.

Related information

Collect PowerFlex management platform log bundle
Retrieve Dell PowerEdge RAID Controller logs

Collect PowerFlex management platform log bundle

You can collect PowerFlex management platform log bundle if the central log collection in PowerFlex Manager is not available or if you are troubleshooting PowerFlex management platform installation problems.

Prerequisites

Retrieve the `pfmp_support` script. In PowerFlex 4.6 or higher, you can obtain the script from the troubleshooting (tsb) pod. Run the following commands on all three management virtual machines (MVMs):

```
kubectl cp -n powerflex $(kubectl get pods -n powerflex | grep tsb | awk '{print $1}'):/var/scripts/pythonScript/pfmp_support /tmp/pfmp_support
```

```
sudo -s
```

```
chmod +x /tmp/pfmp_support
```

About this task

- Use a login shell to connect to the management virtual machines (MVMs) as the root user. Otherwise the PATH variable may not include important tools such as kubectl.
- Run the script as the root user. Otherwise the script may not include important debugging files that non-root users cannot access.

The `pfmp_support` script has a list of all optional arguments. Run `pfmp_support --help --tech` to see the full list. Dell Technologies recommends using these:

`--old-pod-logs`

This argument includes the rsyslog pvc directory which contains historical or exited pod logs. Without this option the resulting log bundle only contains logs from currently running pods, and only the latest of those. This option is necessary for troubleshooting Kubernetes stability issues, upgrade issues, or older problems that are not in current logs.

`--skip-kubernetes-shared`

By default the script collects some kubectl outputs. If the script is run on multiple nodes in the same cluster, the script duplicates the same logs. To save time and space, run the script without this argument on only one node, then use this argument on the remaining nodes.

Steps

1. Run the following command to generate a log bundle on MVM1 that includes Kubernetes logs:

   ```
   sudo --login /tmp/pfmp_support --old-pod-logs
   ```

   You see output similar to the following:

   ```
   estimating required space
   cleaning up temporary directories
   collecting kubernetes data
   collecting shared kubernetes data
   collecting server data
   collecting general hardware data
   collecting network data
   collecting storage data
   preparing files for collection
   generating bundle
   cleaning up temporary directories
   bundle available at '/tmp/powerflex-pfmpsupport/pfmpSupport.tgz'
   ```

2. Run the following command to generate log bundles on MVM2 and MVM3:

   ```
   sudo --login /tmp/pfmp_support --old-pod-logs --skip-kubernetes-shared
   ```

The `--skip-kubernetes-shared` argument skips the Kubernetes logs that were generated on MVM1.

After the log bundles are generated, you must rename each log bundle to include the hostname and change ownership to the **delladmin** user.

3. As the **delladmin** user, run the following command on all three MVMs to rename each bundle to include the hostname:

```
sudo mv /tmp/powerflex-pfmpsupport/pfmpSupport.tgz /tmp/powerflex-pfmpsupport/
$HOSTNAME-pfmpSupport.tgz
```

4. As the **delladmin** user, run the following command on all three MVMs to change file ownership:

```
sudo chown -R delladmin /tmp/powerflex-pfmpsupport
```

Changing file ownership enables WinSCP to access the directory as the **delladmin** user and download the log bundle.

5. Use WinSCP to copy the log bundles from the MVMs.

Related information

Retrieving additional PowerFlex logs

Retrieve Dell PowerEdge RAID Controller logs

Use the following procedure to retrieve the PowerEdge RAID Controller (PERC) logs from ESXi- based and Linux-based servers in a system.

Prerequisites

Ensure that you have access to the following:

- IP address of the node
- Password for the root user in ESXi-based and Linux-based systems
- The PERC utility, PERCCLI, must be enabled. For details, see the following:

  Dell PowerEdge RAID Controller Command Line Interface Reference Guide for your PERC series

  PowerEdge: How to Use the PowerEdge Expandable RAID Controller Command-Line Interface Utility

Steps

1. SSH or RDP to the node.
2. Retrieve the controller information:

Operating System	Run this command

ESXi	`/opt/lsi/perccli/perccli /call show all > <file_name>`
	Example:
	`/opt/lsi/perccli/perccli /call show all > /var/tmp/store/RAIDinfo.txt`

Linux	`/opt/MegaRAID/perccli/perccli64 /call show all > <file_name>`
	Example:
	`/opt/MegaRAID/perccli/perccli64 /call show all > /var/tmp/store/RAIDinfo.txt`

3. Retrieve the `Show Events` log file:

Operating System	Run this command

ESXi	`/opt/lsi/perccli/perccli /call show events file=<file_name>`
	Example:
	`/opt/lsi/perccli/perccli /call show events file=/var/tmp/store/RAIDevents.txt`

Linux	`/opt/MegaRAID/perccli/perccli64 /call show events file=<file_name>`
	Example:
	`/opt/MegaRAID/perccli/perccli64 /call show events file=/var/tmp/store/RAIDevents.txt`

You can retrieve the `RAIDevents.txt` file from your local drive.

4. Retrieve the `Termlog` log file:

Operating System	Run this command

ESXi	`/opt/lsi/perccli/perccli /call show termlog file=<file_name>`
	Example:
	`/opt/lsi/perccli/perccli /call show termlog file=/var/tmp/store/RAIDtermlog.txt`

Linux	`/opt/MegaRAID/perccli/perccli64 /call show termlog file=<file_name>`
	Example:
	`/opt/MegaRAID/perccli/perccli64 /call show termlog file=/var/tmp/store/RAIDtermlog.txt`

You can retrieve the `RAIDtermlog.txt` file from your local drive.

Related information

Retrieving additional PowerFlex logs

Retrieve system event logs from servers

Perform this procedure to access the system events logs from an iDRAC in a PowerEdge server.

Prerequisites

Ensure that you have the iDRAC IP address and port number.

Steps

1. From a web browser, go to **`https://<iDRAC_IP_address:port>`**.
2. In the **Console Login** window, type the user name and password, then click **Login**.
3. In the iDRAC web interface, go to **Overview** > **Server** > **Logs**.

The **System Event Log** page displays a system health indicator, a time stamp, and a description for each event logged. For more information, see the *iDRAC Online Help*.

4. To save the system event logs, click the **Save As** button at the bottom of the table.

   The system event logs are saved as a `.CSV` file.

   NOTE:

   The power supply- and fan-related errors in system event logs may be intermittent in nature, and therefore may display repetitive events for one module.

   This might be a connection-related issue. For such errors, it is recommended that you remove the module and replace it in its socket.

Related information

Retrieving additional PowerFlex logs

Generating a troubleshooting bundle

You can generate a troubleshooting bundle which is a compressed file that contains logging information for PowerFlex Manager managed components.

About this task

If necessary, download the bundle and send it to Dell Technologies Support for issue debugging.

The troubleshooting bundle includes the following logs:

- ASM deployer
- iDRAC life cycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods
- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes
- PowerFlex Block logs
- PowerFlex data server and device gateway target logs

NOTE: You can also generate troubleshooting bundles from the **Resource Groups** page.

Steps

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Generate Troubleshooting Bundle**.
3. Select **Include PowerFlex Block Core Dump logs** to include core dump logs.
4. Select a **PowerFlex Log Level**:
   - Default Node Logs
   - Default Node Logs plus additional MDM information
   - Latest Logs only (Most recent copy of all logs)
5. Select a **Node Selection**:
   - Logs from all nodes
   - Select Specific Nodes

   If you select the **Select Specific Nodes** option and select the number of nodes for which you want to generate the log, the **View/Select Nodes** button is displayed. Click the button to view the list of nodes in the **Node Selection** window. Select the required nodes from the **Available Nodes** list and click **>>** to view them in the **Selected Nodes** list. Click **Save** to return to the **Generate Troubleshooting bundle** page.

   For the **Select Specific Nodes** option **Generate** is enabled only if a node is selected in the **Node Selection** window.

6. Choose a **Delivery Option**:

Option	Description
Send to Configured Dell Technologies connectivity services	If you are using Dell Technologies connectivity services, **Send to Configured Dell Technologies connectivity services** is selected by default. Leave this default setting. If Dell Technologies connectivity services is not configured, this option is disabled.
Download to Share	Provide a path using the following format with credentials: `CIFS:\ \1.1.1.1\uploadDirectory`. Enter a username and password, and click **Test Connection** to verify the connection to the CIFS share before generating the bundle.
Download Locally	Download the troubleshooting bundle to a local file: `\\IP Address\Any folder`

7. Click **Generate**.

Next steps

Sometimes, the troubleshooting bundle does not include log information for all the nodes. The log collection may appear to succeed, but the log for one or more of the nodes may be missing. You may see an error message in the scaleio.trace.log file that says `Could not run get_info script`. If you see this message, you must generate the troubleshooting bundle again to include information for all the logs.

13

Enabling audit logging

This section describes how to enable audit logging in PowerFlex Manager. The procedures in this section explain how to enable logging for PowerFlex events and Ingress audit messages so this information can be forwarded to an external Security Information and Event Manager (SIEM).

Define the PowerFlex events notification policy

Use this procedure to define a notification policy to forward events in the PowerFlex system to the rsyslog-forwarder (also known as the syslog-listener). Then, the rsyslog-forwarder forwards the events to the external destinations that are defined in the policy.

About this task

For this policy, you do not need to define a source, since the required source for PowerFlex events is a built-in feature.

Steps

1. Add a destination:

   First, you must add the identified Security Information and Event Manager (SIEM) server as a destination.

   a. Log in to PowerFlex Manager.
   b. Go to **Settings** > **Events and Alerts** > **Notification Policies**.

      You can also use the following REST API: `dispatch-destinations/post`
   c. From the **Destinations** pane, click **Add**.
      The **Create New Destination Protocol** window opens.
   d. Enter the destination name and description.
   e. From the **Destination Type** menu, select **Syslog**.
   f. Click **Next** and enter the IP, port, and protocol (TCP) of the target SIEM. Ensure that the SIEM IP, port, and protocol are reachable.
2. Create a new policy:

   The new policy defines the rules for processing PowerFlex event messages from sources and specifies to which destination that information should be sent.

   a. Log in to PowerFlex Manager.
   b. Go to **Settings** > **Events and Alerts** > **Notification Policies**.

      You can also use the following REST API: `dispatch-policies/post`
   c. Click **Create New Policy**.
   d. Enter a name and a description for the notification policy. For the policy name, you can enter: **`Powerflex events to external Syslog`**
   e. Set the **Source Type** to **Powerflex_events**.
   f. From the **Resource Domain** menu, select the resource domain for the notification policy. The resource domain options are:
      - All
      - Management
      - Block (Storage)
      - Compute (Servers, Operating Systems, virtualization)
      - Network (Switches, connectivity etc.)
      - Security (RBAC, certificates, CloudLink etc)
   g. Select the checkbox beside the severity levels that you want to associate with this policy.

      The severity indicates the risk (if any) to the system, in relation to the changes that generated the event message.

h. Select the destination that is created in the previous step and click **Submit**.

Define the ingress notification policy

Use this procedure to define a notification policy to forward Ingress audit messages. The purpose of this policy is to capture POST, PUT, and DELETE requests of signed-in users passing through the Ingress Controller and send them to the rsyslog-forwarder, also known as the syslog-listener. The rsyslog-forwarder then forwards them to the external destinations.

Steps

1. Add a syslog source, if you do not have one already:
   a. Log in to PowerFlex Manager.
   b. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
      You can also use the following REST API: `dispatch-sources/post`
   c. From the **Sources** pane, click **Add**.
      The **Add Source** window opens.
   d. Enter a source name and description. For the name, you can enter **`Ingress`**.
   e. For the type, select **Syslog** and click **Enable Syslog**.
2. Create a destination, if you do not have one already.
   You can use the destination that you created to define the event to syslog audit notification policy from the previous procedure.
3. Create a new policy:
   a. Log in to PowerFlex Manager.
   b. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
      You can also use the following REST API: `dispatch-policies/post`
   c. Click **Create New Policy**.
   d. Enter a name and a description for the notification policy. For the policy name, you can enter: **`Powerflex Ingress to external Syslog`**
   e. Set the **Source Type** to **Syslog**.
   f. Set the **Facility** to **Auditlog**.
   g. Select the check box beside the severity levels that you want to associate with this policy.
      The severity indicates the risk to the system in relation to the changes that generated the audit messages.
   h. Select the destination that is created in the previous step and click **Submit**.

Change PowerFlex Manager ingress setting to emit audit messages

Use this procedure to ensure that requests of signed-in users are sent.

About this task

This procedure provides the steps that are required to turn the silent flag off (set it to false) in the Ingress audit plug-in. If this flag is not turned off, the audit messages that are related to POST, PUT, and DELETE requests of signed-in users passing though the Ingress Controller will not be sent.

Steps

1. Log in to one of the management virtual machine nodes with kubectl permissions. Then, copy and paste the following script for enabling or disabling the silent flag:

```
kubectl get cm -n kube-system lua-audit-conf -o yaml > ./lua-audit-conf.current.yaml
sed 's/"silent": true/"silent": false/g'  ./lua-audit-conf.current.yaml > ./lua-audit-
conf.silent-off.yaml
sed 's/"silent": false/"silent": true/g'  ./lua-audit-conf.current.yaml > ./lua-audit-
conf.silent-on.yaml
```

```
echo "Copy + Paste either:"
echo "kubectl apply -f ./lua-audit-conf.silent-on.yaml --force"
echo "OR:"
echo "kubectl apply -f ./lua-audit-conf.silent-off.yaml --force"
```

2. To start the audit, run the following command:

```
kubectl apply -f ./lua-audit-conf.silent-off.yaml --force
```

3. Wait one minute.
4. To ensure the configuration changes made are reflected across all nodes, you must now refresh the user interface. To refresh the user interface, while the browser is in focus, quickly press **F5** five times or press **Ctrl+R** five times.

Results

All Ingress audit messages are now forwarded to the configured SIEM servers.

14

NVMe over TCP

PowerFlex administrators can implement NVMe over TCP and use NVMe over TCP features.

NVMe over TCP load balancing

PowerFlex supports load balancing with NVMe over TCP. NVMe over TCP load balancing requires persistent discovery and host network awareness. You must also add NVMe targets and create a system data network.

Implementing NVMe over TCP load balancing workflow

The following steps summarize the workflow that you must follow to use NVMe over TCP load balancing:

1. Set up persistent discovery on the host operating system.
2. Add NVMe targets.
3. Create a subnet.
4. Create a system data network.

Related concepts

Persistent discovery
NVMe targets
NVMe over TCP host network awareness
Managing system data networks in PowerFlex Manager
Managing system data networks using SCLI

Persistent discovery

The persistent discovery controller ensures that the host remains connected to the discovery service after discovery. If there is a change in the discovery information that is provided to the host, the discovery controller returns an asynchronous event notification (AEN). The host requests the updated **Discovery log** page.

Here are some examples of changes in discovery information:

- A new volume is mapped to the host from a new protection domain.
- A new storage data target (SDT) is added to the system.
- Load balancing wants to move the host connection from one storage data target to another.

NOTE: To use persistent discovery, ensure the following:

- The host operating system supports the Persistent Discovery Controller
- Every host is connected to a subnet to enable discovery
- The persistent flag is used in the NVMe discover command

See the host operating system's documentation for NVMe over TCP host configuration instructions.

Related concepts

NVMe over TCP load balancing
NVMe targets
NVMe over TCP host network awareness
Managing system data networks in PowerFlex Manager
Managing system data networks using SCLI

NVMe targets

NVMe targets (or SDT components) must be configured on the PowerFlex system side, in order to use NVMe over TCP technology.

The NVMe target (or SDT component) is a frontend component that translates NVMe over TCP protocol into internal PowerFlex protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the PowerFlex system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.

TCP ports, IP addresses, and IP address roles must be configured for each NVMe target (or SDT component). You can assign both storage and host roles to the same target IP addresses. Alternatively, assign the storage role to one target IP address, and add another target IP address for the host role. Both roles must be configured on each NVMe target.

- The host port listens for incoming connections from hosts, over the NVMe protocol.
- The storage port listens for connections from the MDM.

Once the NVMe targets have been configured, add hosts to PowerFlex, and then map volumes to the hosts. Connect hosts to NVMe targets, preferably using the discovery feature.

On the operating system of the compute nodes, NVMe initiators must be configured. Network connectivity is required between the NVMe targets and the NVMe initiators, and between NVMe targets (or SDT components) and storage nodes.

Related concepts

NVMe over TCP load balancing
Persistent discovery
NVMe over TCP host network awareness
Managing system data networks in PowerFlex Manager
Managing system data networks using SCLI

Related tasks

Add an NVMe target
Modify an NVMe target
Remove an NVMe target

Add an NVMe target

Add an NVMe target (or SDT component).

Steps

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. Click **Add NVMe Target**.
3. From the **Add NVMe Target** dialog box, configure the following settings:
   a. For **Target Name**, enter a name for the NVMe target (or SDT component).
   b. For **Protection Domain**, select a protection domain.
   c. Accept the default TCP ports, or modify them if necessary.
      - **Storage Port** listens for incoming connections from the MDM, and is only open on an IP address configured with the storage role. Default=12200
      - **I/O Port** listens for incoming connections for I/O, and is only open on an IP address configured with the host role. Default=4420 (which conforms to the NVMe over TCP standard)
      - **Discovery Port** listens for incoming discovery connections, and is only open on an IP address configured with the host role. Default=8009 (which conforms to the NVMe over TCP standard. Use zero to disable the discovery service.)
   d. For **Target IPs**, enter an IP address, and then select a role for the IP address from the **Target IPs Roles** menu. Click **Add Target IP**, and repeat this step until all IP addresses are configured.
4. Click **Add**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Related concepts

NVMe targets

Related tasks

Modify an NVMe target
Remove an NVMe target

Modify an NVMe target

Modify the configuration of an NVMe target (or SDT component).

Steps

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Modify**.
3. In the **Modify NVMe Target** dialog box, modify the desired fields.
4. Optionally, add more target IP addresses, by clicking **Add Target IP**.
5. Click **Modify NVMe Target**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Related concepts

NVMe targets

Related tasks

Add an NVMe target
Remove an NVMe target

Remove an NVMe target

Remove an NVMe target (or SDT component).

Steps

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Remove**.
3. In the **Remove NVMe Target** dialog box, verify that you are removing the wanted NVMe target, and click **Remove**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Related concepts

NVMe targets

Related tasks

Add an NVMe target
Modify an NVMe target

NVMe over TCP host network awareness

Hosts are connected to the storage through a Layer 2 or Layer 3 network. While the storage does not manage the network and is not aware of the network configuration, the storage is impacted by the following network aspects:

- Load balancing takes the data network or subnet into consideration to ensure that there is a balance between the host connections on each data network with a specific subnet for improved performance.
- In Layer 3 networks, the PowerFlex system must have configured routing tables.

Two network objects must be defined for load balancing to work:

Host subnet	Networks or subnets that are used to connect hosts to storage. A subnet is also used to support static routes for data networks. These networks or subnets can be either Layer 2 or Layer 3. Layer 3 networks require configured routing tables. **NOTE:** When configuring NVMe hosts, ensure that every host is connected to a subnet to allow for discovery.
System data network	System-wide object that applies to all protection domains. After a resource group is deployed, you must configure system data networks or subnets to be used to connect the hosts initiator to the storage data target. The number of system data network that you need depends on the number of PowerFlex data networks that are configured for the deployment. PowerFlex data networks are used for storage node to storage node and storage node to SDT communication. You may have to define the system data networks before starting the PowerFlex deployment. **NOTE:** PowerFlex supports a maximum of eight system data networks.

If you have not defined the system data network, by default, a host can reach all system data networks. Not defining the system data networks or subnets may cause the following problems:

- Unequal load between the host initiator ports and nonoptimized I/O performance with the PowerFlex system.
- Impacted path resiliency if some of the host initiator ports cannot connect to the PowerFlex system.

Related concepts

NVMe over TCP load balancing
Persistent discovery
NVMe targets
Managing system data networks in PowerFlex Manager
Managing system data networks using SCLI

Managing system data networks in PowerFlex Manager

The NVMe over TCP load balancer uses the data networks for balancing the host connections. This section summarizes the steps that you must perform in PowerFlex Manager to add, rename, or remove system data networks.

Table 4. Add, rename, or remove data networks in PowerFlex Manager

If you want to...	Do this in PowerFlex Manager
Add system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Enter the required information and click **Save**.
Rename system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Select the network and click **Rename**. 4. Rename the network and click **Apply**.
Remove system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Select the network and click **Remove**. 4. Click **Remove** to remove the network.

Related concepts

NVMe over TCP load balancing
Persistent discovery
NVMe targets
NVMe over TCP host network awareness

Managing system data networks using SCLI

This section provides the SCLI commands that you use to add, rename, or remove system data networks.

For more information on the PowerFlex SCLI commands, see *Dell PowerFlex and PowerFlex Manager CLI Reference Guide*.

Table 5. Using SCLI to manage data networks

If you want to...	Use this SCLI command
Add system data networks	`scli --add_system_network --network_ip <IP> --network_mask <IP> [--network_name <NAME>] (--host_group_id <ID> | --host_group_name <NAME>)`
Rename system data networks	`scli --rename_system_network (--network_id <ID> | --network_name <NAME>) --new_name <NAME>`
Remove system data networks	`scli --remove_system_network (--network_id <ID> | --network_name <NAME>)`

Related concepts

NVMe over TCP load balancing
Persistent discovery
NVMe targets
NVMe over TCP host network awareness
Managing system data networks in PowerFlex Manager
Managing host groups using SCLI
Managing network sets using SCLI

Managing host groups using SCLI

This section provides the SCLI commands that you must perform to add, modify, rename, remove, or query host groups.

This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

The host group holds a subgroup of existing networks that are accessible to a group of hosts. Each host group is associated with a set of system networks. Each host may be associated with multiple host groups. By default, a host is part of the default host group until you create a new host group and associate the host to the new host group. By default, a host supports all data networks and you do not need to configure the host groups. The maximum number of host groups that are allowed per system is 1024.

Table 6. Managing host groups using SCLI

If you want to...	Use this SCLI command
Add host group	`scli --add_host_group [--host_group_name <NAME>] [(--network_id <ID> | --network_name <NAME>)] [(--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)]`
Modify host group	`scli --modify_host_group (--host_group_id <ID> | --host_group_name <NAME>) [(--network_id <ID> | --network_name <NAME>)]`
Rename host group	`scli --rename_host_group (--host_group_id <ID> | --host_group_name <NAME>) [--new_name <NAME>]`

Table 6. Managing host groups using SCLI (continued)

If you want to...	Use this SCLI command
Remove host group	`scli --remove_host_group (--host_group_id <ID> | --host_group_name <NAME>)`
Assign host to group	`scli --assign_host_to_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>) (--host_group_id <ID> | --host_group_name <NAME>)`
Remove host from group	`scli --remove_host_from_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)`
Query host group	`scli --query_host_group (--host_group_id <ID> | --host_group_name <NAME>)`
Query all host group	`scli --query_all_host_group`

Related concepts

Managing system data networks using SCLI
Managing network sets using SCLI

Managing network sets using SCLI

This section provides the SCLI commands that you need to perform to add, modify, rename, remove, or query network sets.

This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

A network set is a set of networks that are connected to the same switch. This switch is configured for balancing the I/O traffic between access switches by associating system data networks to network set. The load balancer takes care of the distribution of networks over network sets.

For example, if a system is defined with two system data networks (192.168.150.0 and 192.168.151.0), then you could create two network sets and associate 192.168.150.0 with network set 1 and 192.168.151.0 to network set 2.

Each system data network belongs to exactly one network set. If the network was not assigned to a network set, the network will be considered a network set of its own.

Use the default network set and do not manually create the network set. Creating network sets is applicable for a software-only deployment.

Table 7. Managing network sets using SCLI

If you want to...	Use this SCLI command
Add network set	`scli --add_network_set [--network_set_name <NAME>] [(--network_id <ID> | --network_name <NAME>)]`
Modify network set	`scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [(--network_id <ID> | --network_name <NAME>)]`
Rename network set	`scli --rename_network_set (--network_set_id <ID> | --network_set_name <NAME>) --new_name <NAME>`

Table 7. Managing network sets using SCLI (continued)

If you want to...	Use this SCLI command
Remove network set	`scli --remove_network_set (--network_set_id <ID> | --network_set_name <NAME>)`
Query network set	`scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)`
Query all network sets	`scli --query_all_network_set`

Related concepts

Managing system data networks using SCLI
Managing host groups using SCLI

Migrating to NVMe over TCP on VMware ESXi

Use the following guidelines when migrating a VMFS datastore and clusters from SDC to NVMe over TCP.

PowerFlex supports three options for migrating from SDC to NVMe over TCP on VMware ESXi:

Online migration using Storage vMotion (VMFS only)

This method is a standard way of using Storage vMotion to move storage and it also supports switching protocols by migrating to a new datastore. This is the method that is described in this guide.

Offline conversion (VMFS only)

Offline conversion is an option for converting an existing VMFS datastore from SCSI (SDC) to NVMe over TCP without having to copy all the data over the network. This option is covered in the Dell Knowledge Base article SCSI to NVMe VMware VMFS Datastore Offline Migration Steps.

Conversion for clustered applications using RDMs

This option converts existing SDC- and RDM-based application clusters to shared VMDKs on NVMe over TCP datastores. This option is covered in the Dell Knowledge Base article PowerFlex: SDC to NVMe/TCP conversion for clustered applications utilizing RDMs on vSphere.

NOTE:

- Dell Technologies recommends using VMware Storage vMotion to migrate data from a volume that is presented by SDC to one presented by NVMe over TCP.
- Linux environments are not discussed in this guide.

Requirements

See the following VMware vSphere 7.0 product documentation link for information about Storage vMotion requirements and limitations: Storage vMotion Requirements and Limitations.

See the following VMware vSphere 8.0 product documentation link for information about requirements and limitations of VMware NVMe Storage: Requirements for VMware NVMe Storage.

See the following VMware KB article for more details about storage migration (Storage vMotion), with the virtual machine powered on: Migrating virtual machines with Raw Device Mappings (RDMs).

Ensure that you satisfy these requirements before you begin the migration:

- The VMware ESXi version is 8.x or later.
- The PowerFlex version is 5.0 or later.

Workflow

The following steps summarize the workflow that you must follow to migrate from SDC to NVMe over TCP using Storage vMotion:

1. Create and map a new volume of equal or greater size than the current VMFS datastore using NVMe over TCP to the same host.
2. Scan for the newly mapped volume.
3. Create a datastore on the NVMe over TCP volume.
4. Perform the standard data migration using Storage vMotion by using a nondisruptive process.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Before migrating the data, you must prepare the VMware ESXi node for mapping NVMe over TCP volumes.

Ensure that you satisfy these requirements before you begin:

- You must have an NVMe over TCP target supported PowerFlex system (version 4.5.x or later).
- Deployed VMware ESXi compute-only nodes.
- The host must be at VMware ESXi version 8.x or later.

Related concepts

Migrating to NVMe over TCP on VMware ESXi

Related tasks

Enable the NVMe over TCP VMkernel ports
Add NVMe over TCP software storage adapter
Copy the host NQN
Add a host to PowerFlex
Create a volume
Map a volume to the host
Discover and connect the NVMe over TCP Target
Perform a rescan of the storage
Create a VMFS datastore on the NVMe over TCP volume

Enable the NVMe over TCP VMkernel ports

Use this procedure to enable the NVMe over TCP VMkernel ports.

Steps

1. Log in to the VMware vSphere Client.
2. Click **Home/Inventory** and select the host.
3. Select **Configure** > **VMkernel adapters**.
4. Edit **PowerFlex-Data 1**.
5. Select the **NVMe over TCP** check box and click **OK**.

6. Repeat these steps for the remaining PowerFlex data networks.
7. Repeat the steps for the remaining hosts in the cluster.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Add NVMe over TCP software storage adapter

Use this procedure to add an NVMe over TCP software storage adapter.

Steps

1. Log in to the VMware vSphere Client.
2. Click **Home** > **Inventory** > **Hosts and Clusters**.
3. Select the cluster of PowerFlex compute-only nodes that you want to configure.
4. From the right pane, click **Configure** > **Storage Adapters**.
5. From the right pane, click **Add Software Adapter**.
6. Click **Add NVMe over TCP adapter**.
7. Select the first VMNIC and click **OK**.
8. Click **Add NVMe over TCP adapter**.
9. Select the second VMNIC and click **OK**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Copy the host NQN

Use this procedure to copy the host NQN to the copy buffer. The host NQN details are required when you add the host to PowerFlex.

Steps

1. Log in to VMware vSphere Client.
2. Select the first host.
3. From the right pane, click **Configure** > **Storage Adapters**.
4. Select the first VMware NVMe over TCP storage adapter. For example, **vmhba6x**.
5. From the pane, select **Controllers/Add Controller**.

   The host NQN is listed at the top of the form.
6. Click **COPY** and place the host NQN in the copy buffer.
7. Click **CANCEL**.
8. Repeat the steps for all the hosts.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Add a host to PowerFlex

Use this procedure to add a host to PowerFlex.

Steps

1. Log in to PowerFlex Manager.

2. Click **Block** > **Hosts**.
3. Click **Add Host**.
4. Enter the hostname and paste the host NQN from the copy buffer.
5. Enter the **Number of Paths Per Volume**. The default number of paths is four per volume, and the maximum number of paths is eight per volume.
6. Enter the **Number of System Ports per Protection Domain**. The default number of system ports is ten per protection domain, and the maximum number of system ports is 16 per protection domain.
7. Click **Add**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Create a volume

Use this procedure to create a volume.

Steps

1. Log in to PowerFlex Manager,
2. Click **Block** > **Volumes**.
3. Click **Create Volume**.
4. Enter the number of volumes and the name of the volumes.
5. Enter the required volume size in GB, specifying the size in 1 GB increments.
6. Select the NVMe storage pool and click **Create**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Map a volume to the host

Use this procedure to map a volume to the host.

Steps

1. Log in to PowerFlex Manager,
2. Click **Block** > **Volumes**.
3. Select the volume check box and click **Provision** > **Map**.
4. Select the protocol **NVMe**.
5. Select the check box for the host to which you are mapping the volume.
6. Click **Map**.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Discover and connect the NVMe over TCP Target

Use the procedure to discover and connect the NVMe over TCP Target PowerFlex system. Use the `esxcli` command to perform the operation.

Steps

1. Log in to the ESXi server using ssh.

2. Run the discovery query on each adapter:

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.x.y -p 8009
```

In the first example above, 6x is the first NVMe over TCP software adapter and 6y is the second NVMe over TCP software adapter.

In the second example above, 192.168.x.x is first data IP Address and 192.168.x.y is second data IP Address, depends on which VMNIC is enabled for which software NVMe over TCP adapter.

3. Connect to the PowerFlex system by appending "-c" to the discovery query command.

```
#esxcli nvme fabrics discover -a vmhba64 -i 192.168.x.x -p 8009 -c
#esxcli nvme fabrics discover -a vmhba65 -i 192.168.x.y -p 8009 -c
```

4. Get the controller list by verifying the connected controllers.

```
#esxcli nvme controller list
```

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Perform a rescan of the storage

Use this procedure to perform a rescan of the storage.

Steps

1. In the vSphere Client object navigator, browse to a host, a cluster, a data center, or a folder that contains hosts.
2. From the right-click menu, select **Storage** > **Rescan Storage**.
3. Specify the extent of the rescan.

Option	Description
Scan for New Storage Devices	Rescan all adapters to discover new storage devices. If new devices are discovered, they appear in the device list.
Scan for New VMFS Volumes	Rescan all storage devices to discover new datastores that have been added since the last scan. Any new datastores appear in the datastore list.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Create a VMFS datastore on the NVMe over TCP volume

Use this procedure to create a VMFS datastore on the NVMe over TCP volume.

Steps

1. Log in to VMware vCenter.
2. Select **Home/Inventory** and select the storage icon.
3. Right-click the cluster of PowerFlex compute-only nodes and select **Storage/New Datastore**.
4. Select VMFS and click **NEXT**.
5. Enter the name of the datastore and select a host from the list.
6. You should see a new disk called NVMe TCP Disk (`eui.######`).
7. Select the disk and click **NEXT**.
8. Select VMFS 6 and click **NEXT**.

9. Leave the default configuration unless you have been otherwise instructed and click **NEXT**.
10. Review your selections and click **FINISH**.

Results

You are ready to migrate the data with Storage vMotion.

Related concepts

Prepare the VMware ESXi node for mapping NVMe over TCP volumes
Migrate the data with Storage vMotion

Migrate the data with Storage vMotion

After you prepared the node, you can migrate the data with Storage vMotion.

Follow the standard VMware vSphere 7.0 procedure to migrate the virtual machines from an SDC-based datastore to an NVMe over TCP-based datastore: Migrate a Virtual Machine to New Storage

Related tasks

Create a VMFS datastore on the NVMe over TCP volume

15

Migrate VMFS datastores from SDC to NVMe-oF

You can migrate VMFS6 datastores from SCSI-controlled storage data clients (SDCs) to Non-Volatile Memory Express over Fabrics (NVMe-oF) by using an offline migration method.

CAUTION: These instructions apply only to VMFS6 datastores.

Offline VMFS datastore migration from a SCSI-controlled SDC to NVMe-oF does not involve data movement. The device previously presented to a VMware ESXi host or cluster as an SDC device is unpresented and presented again as an NVMe-oF device. The VMFS datastore is then signatured again and made available to hosts, and retains all VM contents.

Migration workflow

The migration process happens in three phases: pre-migration, migration, and post-migration. The following table provides a migration workflow checklist.

Table 8. Migration workflow checklist

Pre-migration	Migration	Post-migration
Pre-migration includes preparatory checks and validation. 1. Check feature impact 2. Verify VMFS locking mode is `ATS` and not `ATS+SDC (SCSI)` 3. Check for RDM usage on VMs 4. Review claim rules and settings 5. Verify device and path limits 6. Check for unsupported features 7. Assess feature impact post-migration	Migration is the actual migration process. 1. Power off and unregister all VMs 2. Unmount VMFS volume from all hosts 3. Check VMFS metadata consistency 4. Detach SDC volume from ESXi hosts 5. Stop presenting SDC volume from array 6. Present device as NVMe-oF to one ESXi host 7. Rescan adapters on the host 8. Check VMFS metadata consistency after conversion 9. Re-signature the VMFS volume 10. (Optional) Rename the datastore 11. Check VMFS metadata consistency after re-signature 12. Present device as NVMe-oF to all ESXi hosts 13. Register and power on all VMs	Post-migration includes using vSphere Client or another interface to verify that the VMs are successfully running.

Additional resources and support

Using vSphere On-disk Metadata Analyzer (VOMA): Checking Metadata Consistency with vSphere On-disk Metadata Analyzer.

Contact VMware Support if you encounter issues during migration. Consult your storage array vendor's documentation for array-specific procedures.

Offline migration feature impact

Offline migration impacts the features listed in the following tables.

CAUTION: Review the following tables to understand which features are unavailable and which perform differently after migration.

Unsupported features

The following features are not available after offline migration.

Table 9. Unsupported features

Feature	Description	Notes
Guest clustering	Clustered virtual machine disk (VMDK) feature supporting high availability solutions such as Windows Server Failover Cluster (WSFC)	A VMFS datastore with clustered VMDK enabled cannot be migrated
Site Recovery Manager (SRM)	Array-based replication with SRM is not supported by NVMe-oF	Migrating datastores involved with SRM array replication renders the solution useless

Performance impact on supported features

The following features are available but perform differently after offline migration.

Table 10. Performance impact on supported features

Feature	Nature of impact	Action required
Hardware Accelerated Move (XCOPY)	No equivalent command to XCOPY. VMware software data mover is used instead. This may decrease performance of operations like cloning or SvMotion.	None
WRITE SAME command with the UNMAP bit	If NVMe-oF device does not support NVMe-oF equivalent of write zeroes or unmap, there may be performance impact.	None

Complete pre-migration checks

You must complete checks to understand environment characteristics and features in use. These checks determine if offline migration is possible and help you understand the post-migration impact.

Prerequisites

Before you begin, ensure the following:

- The ESXI hosts must be configured for NVMe-oF.
- The Storage Data Target (SDT) is configured in PowerFlex.

CAUTION:

- **All VMs in the VMFS datastore must be powered off prior to starting the migration.**
- **The volume must support ATS mode. Only attempt migration if the VMFS6 datastore uses ATS-only locking mode and does not use SDC SCSI-2 reservations.**

Steps

1. Check the VMFS volume locking mode. Run the following command to check the locking mode:

```
esxcli storage vmfs lockmode list -l <volume name/label>
```

Example output for a volume that supports offline migration:

```
Volume Name UUID                                    Type    Locking Mode ATS Compatible ATS
Upgrade Modes ATS Incompatibility Reason
----------- ------------------------------------ ------ ------------ --------------
----------------- -------------------------
testVol1    5d1c5b0f-xxxxxxxx-xxxx-246e9xxxxdb0 VMFS-6 ATS          true           No
upgrade needed
```

CAUTION:

- **Offline migration is supported only if the locking mode is `ATS`. The `ATS+SDC (SCSI)` mode is not supported.**
- **Local drives on the targeted server may respond with `Device does not support ATS`. You can still perform offline migration if this response is returned.**

Example output for a volume that does NOT support offline migration:

```
Volume Name UUID                                    Type       Locking Mode       ATS
Compatible    ATS Upgrade Modes    ATS Incompatibility Reason
----------- ------------------------------------ ------ ------------ --------------
----------------- -------------------------
testVol2    63510e51-xxxxxxxx-xxxx-246e9xxxxde6   VMFS-6    ATS+SDC (SCSI)      false
   None              Device does not support ATS
```

2. Check for Raw Device Mapping (RDM) usage. Use the Dell Virtual Storage Integrator (VSI) plug-in to view disk types for each VM. The plug-in lists VMs by Runtime Name and notes if the VM has an RDM.

   NOTE: If a VM in the selected datastore has an RDM, the VM cannot be migrated to NVMe-oF.

   Do ONE of the following if you find an RDM:

   - Remove the RDM from the VM.
   - Convert the RDM.
   - Move the VM to another datastore before migration.

3. Check the claim rules and settings. Custom claim rules on the SDC device that exist before migration are not applied when the device is presented through NVMe-oF.
   - NVMe-oF devices do not present separate vendor and model fields through inquiry.
   - Fields are combined, which requiring a new claim rule if needed.
   - Claim rules based on device identifiers, such as World Wide Name (WWN), will fail because SDC and NVMe-oF identifiers are distinct.
   - By default, VMware claims newly presented NVMe-oF devices with the default pathing in the High-Performance Plug-in (HPP).

4. Check the device and path limits.

   NVMe-oF supports fewer devices and paths per ESXi host than SDC.

Protocol	Devices per Host	Paths per Host
SDC	1024	4096
NVMe-oF	256	2048

   If SDC devices exceed NVMe-oF limits, do the following:

   - Employ more ESXi hosts.
   - Use VMware Storage vMotion to consolidate datastores before or after conversion.

5. Check for unsupported features. Some VMware features are not supported when you use NVMe-oF. Before migration, check feature support in your VMware ESXi version's release notes on the Broadcom TechDocs site.
6. Check for feature impact post-migration. The following features may have performance or functional changes when using NVMe-oF compared to SDC:

Feature	Nature of Impact	Action to be taken
WRITE SAME command with the UNMAP bit	If an NVMe-oF device does not support the NVMe-oF equivalent of write	None

Feature	Nature of Impact	Action to be taken
	zeroes or unmap, there could be a performance impact.	

Power off VMs and unmount volumes

Power off and unregister all VMs hosted on the datastore that you are migrating. Then, unmount the VMFS volumes from all hosts.

Prerequisites

Ensure that you have a list of the VMs in the datastore and the folder where the `VMX` file resides so you can register the VMs again after migration.

About this task

CAUTION: Do not delete the VMs, only unregister them.

Steps

1. To power off and unregister all VMs and templates, log in to vSphere Client.
2. Right-click the VM.
3. Select **Power** > **Power Off**.
   If VMware Tools is installed, Dell Technologies recommends using **Shut Down Guest OS**.
4. Click **Yes** to confirm.
5. Right-click the VM again.
6. Click **Remove from Inventory**.
7. Click **Yes** to confirm.

Unmount the VMFS volume from all ESXi hosts to ensure the volume is not in use during the consistency check and during migration. Do the following in vSphere Client:

8. Go to **Storage**.
9. Select the datastore.
10. Right-click the datastore.
11. Click **Unmount Datastore**.
12. On the **Unmount Datastore** page, select the hosts.
13. Click **OK**.

Check VMFS volume metadata consistency

Before you initiate migration, check the consistency of VMFS on-disk metadata to ensure no inconsistencies exist.

Prerequisites

The VMs are powered off and unregistered, and the VMFS volumes are unmounted.

About this task

Check the consistency of VMFS on-disk metadata and correct any inconsistencies.

Steps

1. Run VMware On-Disk Metadata Analyzer (VOMA) in check mode:

```
voma -m vmfs -f check -d /vmfs/devices/disks/<DEVICE>:<PARTITION> -s <OUTPUT_FILE>
```

In the previous command:

- *<DEVICE>* is the SDC device hosting the VMFS6 volume being migrated.
- *<PARTITION>* is the partition number on which the VMFS volume is formatted.
- *<OUTPUT_FILE>* is the absolute path of the file to save command output. This can be in /tmp or any VMFS volume other than the one being migrated.

For example:

```
voma -m vmfs -f check -d /vmfs/devices/disks/naa.60000970000120200302533030313031:1
-s /tmp/voma.out
```

Example output:

```
voma -m vmfs -f check -d /vmfs/devices/disks/eui.c70fd6f05a6ab90fc01b0dc500000005:1
-s tmp/voma.out
```

```
Running VMFS Checker version 2.1 in check mode
Initializing LVM metadata, Basic Checks will be done
Checking for filesystem activity
Performing filesystem liveness check..|Scanning for VMFS-6 host activity (4096 bytes/HB,
1024         SDC (SCSI) 2 reservation successful
             \
Phase 1: Checking VMFS header and resource files
   Detected VMFS-6 file system (labeled:'SDS2-storage') with UUID:69eafa45-51188a9f-
a0ac-58a2e139218c, Version 6:82
Phase 2: Checking VMFS heartbeat region
Phase 3: Checking all file descriptors.
Phase 4: Checking pathname and connectivity.
Phase 5: Checking resource reference counts.
Total Errors Found:           0
```

NOTE: If the command returns the error `VOMA failed to check device: Device or Resource busy`, the VMFS is not properly unmounted.

2. Analyze the output file for metadata inconsistencies. If you find inconsistencies, address them by running VOMA in advanced fix mode before continuing.

```
voma -m vmfs -f fix -d /vmfs/devices/disks/<DEVICE>:<PARTITION>
```

For example:

```
voma -m vmfs -f fix -d naa.60000970000120200302533030313031:1
```

```
Running VMFS Checker version 2.1 in fix mode
Initializing LVM metadata, Basic Checks will be done
Checking for filesystem activity
         SDC (SCSI) 2 reservation successful                        st activity (4096
bytes/HB, 1024 HBs).
Phase 1: Checking VMFS header and resource files
   Detected VMFS-6 file system (labeled:'SRM_UPGRADE_1') with UUID:6418928f-d0fb0a78-
fa29-34800d0ed39c, Version 6:82
Phase 2: Checking VMFS heartbeat region
Phase 3: Checking all file descriptors.
Phase 4: Checking pathname and connectivity.
Phase 5: Checking resource reference counts.
Total Errors Found:           0
Total Errors Fixed:           0
```

Collect the VMFS metadata dump. Save a metadata dump for reference if any metadata inconsistencies are seen in subsequent steps.

For more details on using VOMA in check mode, advanced fix mode, or dump mode, see Checking Metadata Consistency with vSphere On-disk Metadata Analyzer.

Detach and stop presenting the SDC

Detach the SDC device from ESXi hosts, and stop presenting the SDC device from the array.

Prerequisites

You checked the VMFS volume metadata consistency.

Steps

1. Log in to vSphere Client.
2. Select an ESXi host.
3. Under **Configure**, select **Storage** > **Storage Devices**.
4. Right-click the Network Address Authority (NAA) ID of the device.
5. Click **Detach**.
6. In the **Confirm Detach Device** window:
   a. Select **Detach the selected device from multiple hosts**.
   b. Check the listed hosts.
   c. Click **Yes**.

Use the following steps to stop presenting the SDC device that is specific to your storage array. Consult your array-specific documentation for the procedure. The following steps are for PowerFlex Manager:

7. Log in to PowerFlex Manager.
8. On the dashboard, select **Block** > **Volumes**.
9. Select the volume by checking the box.
10. Select **Provision** > **Unmap**.
11. In the **Unmap** pane:
    a. Check the hosts you want to unmap.
    b. Click **Unmap**.

Present and scan the device

Present the device as NVMe-oF to the ESXi host, and initiate device rescan on the host.

Prerequisites

You detached and stopped presenting the SDC.

About this task

The steps to present the device again using NVMe-oF are storage array-specific. Consult your array-specific documentation for the procedure. The following steps are for PowerFlex Manager:

Steps

1. Log in to PowerFlex Manager.
2. On the dashboard, select **Block** > **Volumes**.
3. Select the volume by checking the box.
4. Select **Provision** > **Map**.
5. In the **Map Volumes to Hosts** window:
   a. Click the **NVMe-oF** option.
   b. Check the host or hosts that you want to map.
   c. Click **Map**.

Once the device is presented using NVMe-oF, discovery is typically immediate. If the device does not appear, scan the adapters using vSphere UI or CLI.

6. Log in to vSphere Client.

7. Right-click the cluster.
8. Select **Storage** > **Rescan Storage**.
9. In the **Rescan Storage** window:
   a. Check both options: **Scan for new Storage Devices** and **Scan for new VMFS Volumes**.
   b. Click **OK**.
10. Optional: You can also run this command in the vSphere CLI:

```
esxcli storage core adapter rescan -a
```

Check VMFS volume metadata consistency after conversion

After converting a device to NVMe-oF, check the VMFS metadata consistency again.

Prerequisites

You presented the device as NVMe-oF to the ESXi host.

About this task

On the ESXi host with access to the device, run VOMA in check mode again to validate that VMFS on-disk metadata is still consistent.

CAUTION: VOMA uses the SDC (SCSI-2) reserve command to lock the device. NVMe-oF devices do not support an equivalent of SDC (SCSI-2) reservation. Because of this limitation, you must pass the `-N` option to VOMA when the backend device is NVMe-oF.

Steps

1. Run the following command:

```
voma -m vmfs -f check -N -d /vmfs/devices/disks/<DEVICE>:<PARTITION> -s <OUTPUT_FILE>
```

   The following warning is displayed:

```
########################################################################
# Warning !!!                                                          #
#                                                                      #
# You are about to execute VOMA without device reservation.            #
# Any access to this device from other hosts when VOMA is running      #
# can cause severe data corruption                                     #
#                                                                      #
# This mode is supported only under VMware support supervision.        #
########################################################################
VMware ESXi Question:
Do you want to continue (Y/N)?
0) _Yes
1) _No

Select a number from 0-1:
```

   CAUTION: This warning notifies you that it is your responsibility to prevent the volume from being mounted or accessed concurrently from other hosts while VOMA is running. If you correctly followed the previous steps and the device is mapped and discovered on only one ESXi host, it is safe to proceed.

   Enter **0** at the prompt to continue with VOMA check mode.

   Example command and output:

```
voma -m vmfs -f check -N -d /vmfs/devices/disks/eui.c01b0dc5000000056
```

```
Running VMFS Checker version 2.1 in check mode
Initializing LVM metadata, Basic Checks will be done
```

```
Checking for filesystem activity
         Reservation Support is not present for NVMEOF devices st activity (4096
bytes/HB, 1024
Performing filesystem liveness check..|
###########################################################################
#   Warning !!!                                                           #
#                                                                         #
#   You are about to execute VOMA without device reservation.             #
#   Any access to this device from other hosts when VOMA is running       #
#   can cause severe data corruption                                      #
#                                                                         #
#   This mode is supported only under VMware support supervision.         #
###########################################################################
Do you want to continue (Y/N)?
0) _Yes
1) _No
Select a number from 0-1: 0
Phase 1: Checking VMFS header and resource files
   Detected VMFS-6 file system (labeled:'SDS2-storage') with UUID:69eafa45-51188a9f-
a0ac-58a2
Phase 2: Checking VMFS heartbeat region
Phase 3: Checking all file descriptors.
Phase 4: Checking pathname and connectivity.
Phase 5: Checking resource reference counts.
Total Errors Found:           0
```

2. If you find errors, investigate before proceeding.

Update VMFS volume signature

After a device is presented as NVMe-oF, you must update the signature on the datastore. The current signature is based partly on the World Wide Name (WWN) of the device when presented using SDC. Because the NVMe-oF device ID is different, you must generate a new signature.

Prerequisites

You checked the VMFS volume metadata consistency again.

Steps

1. Log in to vSphere Client.
2. Go to **Storage** and select the cluster.
3. Right-click the cluster.
4. Click **Storage** > **New Datastore**.
5. On the **New Datastore** page:
   a. Select the **VMFS datastore** type.
   b. Click **Next**.
6. Assign a new name to the datastore.
7. Select the host.
8. Select the **NVMe-oF TCP** disk.

   The **Snapshot Volume** column confirms that ESXi has successfully recognized a previous VMFS SCSI signature on this specific disk.
9. Click **Next**.
10. On the **Mount Option** page:
    a. Select **Assign a new signature**.
    b. Click **Next**.
11. Click **Finish**.

Alternately, you can also run these commands in the vSphere CLI:

12. Dell Technologies recommends that you scan the file system again by running the following command:

```
esxcli storage filesystem rescan
```

13. Get a list of VMFS snapshot devices:

```
esxcli storage vmfs snapshot list
```

Check that the newly presented NVMe-oF device is present. Other snapshots unrelated to this process may also appear.

14. Update the VMFS volume signature:

```
esxcli storage vmfs snapshot resignature --volume-label=<label>|--volume-uuid=<id>
```

15. Optional: Rename the VMFS datastore.
When a VMFS volume is re-signatured, the VMFS volume label is prefixed with `snap` followed by an alphanumeric string.

For example, if the original datastore was `sds-storage`, it may now be named `snap-5c42a2bc-sds-storage`.

To revert to the original name:

a. Right-click the newly added datastore. For example, `snap-5c42a2bc-sds-storage`.
b. Click **Rename**.
c. Rename the datastore back to the original name by removing the `snap-` prefix.

Check the VMFS volume metadata consistency after applying the new signature.

16. Validate that the on-disk VMFS metadata is consistent after re-signature. Run the following VOMA command to check the VMFS volume:

```
voma -m vmfs -f check -N -d /vmfs/devices/disks/<DEVICE>:<PARTITION> -s <OUTPUT_FILE>
```

In the previous command:

- *<DEVICE>* is the SDC device hosting the VMFS6 volume being migrated.
- *<PARTITION>* is the partition number on which the VMFS volume is formatted.
- *<OUTPUT_FILE>* is the absolute path of the file to save command output. This can be in /tmp or any VMFS volume other than the one being migrated.

CAUTION: Verify if VOMA reports any inconsistencies. Proceed only if VOMA does not report any errors.

17. If there were no issues in previous steps, present the device using NVMe-oF to all ESXi hosts in the cluster.

NOTE: NVMe-oF devices are recognized immediately. If not visible, rescan adapters through vSphere UI or CLI.

18. Required: For verification purposes, ensure that the VMFS6 volume is mounted and accessible on all hosts.

Register and power on all VMs and templates

Register all VMs hosted on the datastore and power them on. Verify that VMs power on successfully and that you can access the virtual machine disks (VMDKs).

Prerequisites

You updated the VMFS volume signature.

About this task

Register and power on all VMs and templates.

Steps

1. Log in to vSphere Client.
2. Go to **Storage**.
3. Right-click the new datastore.

4. Select **Register VM**.
5. Select the VMs to register.
6. Click **OK**.
7. On the **Register Virtual Machine** page:
   a. Select the datacenter.
   b. Click **Next**.
8. Select the cluster to place the VM.
9. Click **Next**.
10. Click **Finish**.

To power on VMs:

11. Go to **Hosts and Clusters**.
12. Right-click the VM.
13. Click **Power On**.

> **NOTE:** When powering on VMs from the vCenter UI, you may see a dialog box asking if the VM was copied or moved. Select **I Copied it** in the dialog box.

16

Install a storage data client

Use the following procedures to manually install a storage data client on different operating systems.

Installing the SDC manually

If a host of the SDC type was not installed on front-end servers during initial installation activities, follow the appropriate instructions for your front-end servers' operating system.

See the *Dell PowerFlex and PowerFlex Manager Support Matrix* to ensure that the intended SDC operating system of the host is compatible for use with your PowerFlex system.

You can use any of the following methods to install the SDC manually:

- VMware ESXi
- Linux
- Windows

Install the SDC on a VMware VCF/ESXi 9.x based server

Use this procedure to install the SDC on a VMware VCF/ESXi 9.x based server.

About this task

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

NOTE: This procedure requires multiple servers reboots.

Prerequisites

Ensure that you have the following:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this procedure to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host.
- The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- A GUID string for the SDC. These strings can be generated by tools that are available online, for example GUID generator or UUID generator. The GUID must conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

Steps

1. On the VMware ESXi-based server on which you are installing the SDC, set the acceptance level, type the following: **`esxcli software acceptance set --level PartnerSupported`**
2. Put the VMware ESXi node into maintenance mode: **`esxcli system maintenanceMode set --enable true`**
3. Install the SDC agent package: **`esxcli software component apply -d /<PATH>/scaleio-daemon-esx_5.x.x.xxx-x.zip`**

   Where **`<PATH>`** is the directory containing the SDC agent package (.zip file).

4. Install the SDC driver package: **esxcli software component apply -d /<PATH>/scaleio-sdc-esx9x_5.x.x.xxx-xOEM.xxx.x.xxxxxxxx.zip**.
5. Get permission to reboot the VMware ESXi-based server, and perform a graceful reboot cycle to load the SDC driver: **reboot**
6. Ensure that the driver is loaded: **vmkload_mod -l | grep scini**.

   Output similar to the following indicates that the driver is loaded:

   ```
   scini 2 808
   ```

7. Set the module parameters:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
   IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
   ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the virtual IP addresses, or the MDM IP address. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a , symbol.
     - Separate multiple MDM clusters with the + symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one, use the following format for the `IoctlMdmIPStr` parameter: `10.20.30.40,50.60.70.80+11.22.33.44`
   - <XXXXXX> is the user-generated GUID string. Example GUID: `12345678-90AB-CDEF-1234-567890ABCDEF`
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.

   For example:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=a9745569-
   eed1-43fe-82fd-b28199dd7ffb  IoctlMdmIPStr=192.168.211.93,172.168.211.108"
   ```

8. Reboot the VMware ESXi host to set the parameters.
9. Exit the VMware ESXi host from maintenance mode: **esxcli system maintenanceMode set --enable false**
10. Back up the parameters: **/bin/auto-backup.sh**.
11. Log in to the primary MDM or PowerFlex Manager to verify that the SDC is connected, and proceed with volume mapping.

Install the SDC on a VMware ESXi 8.x server

Install the SDC on a VMware ESXi 8.x server using command-line esxcli commands.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host.
- The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID must conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

NOTE: This procedure requires two servers reboots.

Steps

1. On the VMware ESXi-based server on which you are installing the SDC, set the acceptance level:

```
esxcli software acceptance set --level PartnerSupported
```

2. Install the SDC:
   a. Change the directory where you have copied SDC.ZIP file.
   b. Unzip the SDC.ZIP file, run the following command: **`unzip <filename.zip>`**.
   c. Install the SDC package, run the following command: **`esxcli software vib install -v <path_to_vib>`**.

   Where **`<path_to_vib>`** is the path to SDC.VIB file which is unzipped in the above command.

3. Get permission to reboot the VMware ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot are required, as described in this procedure.

4. Set the IP address of the MDM:

```
esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a "," symbol.
     - Separate multiple MDM clusters with the + symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one:

       `10.20.30.40,50.60.70.80+11.22.33.44`
   - *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.

   CAUTION: A few applications require an advanced settings in ESXi for specific vital product data (VPD) pages. For more information about changing `blkScsiUnitSerialLenByStandard` parameter on an ESXi host for PowerFlex volumes, see Procedure to change parameter blkScsiUnitSerialLenByStandard on ESXi for PowerFlex volumes. Perform this procedure only if your applications require this setting. This is not a common setting.

5. Back up the parameters:

```
/bin/auto-backup.sh
```

6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

```
vmkload_mod -l | grep scini
```

   Output similar to the following indicates that the driver is loaded:

```
scini 2 808
```

   The SDC component is installed on the VMware ESXi-based server.

Results

The SDC is installed on the VMware ESXi-based server and is connected to PowerFlex.

Install the SDC on a Linux-based server

Install the storage data client on a Linux-based server with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more storage data clients to an existing system, and for adding storage data clients to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- Secure boot is not supported; ensure that secure boot is disabled in the BIOS.

About this task

This section describes how to manually install an external SDC on a Linux-based server using the command line.

The steps in the procedure apply to Citrix Hypervisor(XenServer)/Red Hat Enterprise Linux/Oracle Linux operating systems.

Steps

1. Install the GPG key on every server on which SDC is installed. From the PowerFlex installation folder, run the following command on every server:

   ```
   rpm --import RPM-GPG-KEY-ScaleIO
   ```

2. From the command line, install the SDC:
   - Red Hat Enterprise Linux/Oracle Linux/SLES

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
     ```

   - Ubuntu (Extract these files before use, as described above.)

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
     ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located
3. Verify that the driver is loaded using one of the following commands:
   - SLES 15:

     ```
     systemctl status scini
     ```

   - All other Linux:

     ```
     service scini status
     ```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Install SDC on Windows-based PowerFlex compute-only node

Use this procedure to install SDC on the Windows-based PowerFlex compute-only node.

Steps

1. Download PowerFlex LIA and the SDC plugin.

2. Double-click the PowerFlex LIA setup file.
3. Accept the terms and license agreement and click **Install**.
4. Click **Finish**.
5. Do the following for Windows-based SDC node before deploying the SDC component using the command prompt with administrator privileges:
   a. Type **bcdedit /debug ON**
   b. Type **bcdedit /**set testsigning ON
   c. Type **reboot windows**
6. If you know the MDMs VIPs before installing the SDC component, run the command:

```
msiexec /i HYPERLINK "http://
vm-sio-fs.xiolab.lab.emc.com/sw_dev/Artifacts/Build-All-WIN16/5.0.0.12015/release/EMC-
ScaleIO-sdc-5.0-0.12015.msi"EMC-ScaleIO-sdc-5.x-x.xxxxx.msi msi
MDM_IP=<LIST_VIP_MDM_IPS>
```

   For example:

```
msiexec /i HYPERLINK "http://
vm-sio-fs.xiolab.lab.emc.com/sw_dev/Artifacts/Build-All-WIN16/5.0.0.12015/release/EMC-
ScaleIO-sdc-5.0-0.12015.msi"EMC-ScaleIO-sdc-5.x-x.xxxxx.msi msi
MDM_IP=192.168.151.185,192.168.152.185
```

7. Accept the terms and license agreement and click **Install**.
8. Click **Finish**.
9. On confirmation from the customer to reboot the Windows server, perform a reboot to load the SDC driver on the server.
10. If the MDMs VIPs are not provided before installing the SDC component:
    a. Double-click the PowerFlex SDC setup.
    b. Accept the terms in the license agreement and click **Install**.
    c. Click **Finish**.
    d. Type **c:\Program Files\EMC\scaleio\sdc\bin>drv_cfg.exe --add_mdm --ip <VIPs_MDMs>** to configure the PowerFlex node in the PowerFlex software.

Mapping volumes to Windows-based PowerFlex compute-only nodes

Use this procedure to map volumes to Windows-based PowerFlex compute-only nodes.

Steps

1. Log in to PowerFlex Manager.
2. Verify that the Windows-based PowerFlex compute-only nodes are listed under **Lifecycle** > **Storage nodes**.
3. Click **Block** > **Volumes**.
4. Select the **Volume** that you want to map.
5. Click **Provision** > **MAP**.
6. Select the newly added Windows host and click **MAP**.
7. Log in to a Windows-based PowerFlex compute-only node and select **Disk Management**.
8. Rescan the disk and bring it online.
9. Click **Initialize disk**.
10. Create **New Simple Volume** and click **Next**.
11. Select **Default** and click **Next**.
12. Assign the drive letter.
13. Select **Default** and click **Next**.
14. Click **Finish**.